$t\bar{t}$ and $hh \to b\bar{b}\,\ell\nu\,q\bar{q}'$: samples, the ten angles, and the $t\bar{t}$ density $f_{t\bar{t}}$ by OSDE as a signal-agnostic discriminant

Internal note

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Abstract

The orthogonal-series density $f_{t\bar{t}}$ of the ten $t\bar{t}$ angles, fitted on $t\bar{t}$ only, separates $hh \to b\bar{b}\,\ell\nu\,q\bar{q}'$ from $t\bar{t}$ at parton level (AUC 0.67) but not after reconstruction (0.53 at particle level, 0.51 at detector level). The loss is traced to the $t\bar{t}$-hypothesis reconstruction, which rebuilds the off-shell hh events as on-shell $t\bar{t}$; the information survives in the resonance masses. Multiplying $f_{t\bar{t}}$ by one Breit–Wigner factor per resonance mass, with no fitted coefficients, recovers the parton-level performance at particle level (0.69) and most of it at detector level (0.60). Only about 20 angular terms matter; realistic quark tagging changes nothing once the masses are in. Comparisons with other unsupervised and with supervised machine-learning methods, a BSM benchmark and a pileup study are included.

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1 Introduction and summary

This note documents a study of the orthogonal-series density estimate (OSDE) $f_{t\bar{t}}(\mathbf{x})$ of the ten $t\bar{t}$ decay and production angles $\mathbf{x} = \{Y, \Theta, \theta, \phi, \theta', \phi', \tilde{\theta}, \tilde{\phi}, \tilde{\theta}', \tilde{\phi}'\}$ defined in the paper, as a signal-agnostic (unsupervised) discriminant: $y = f_{t\bar{t}}(\mathbf{x})$ is low for events that do not look like $t\bar{t}$, whatever they are. The test signal is $hh \to b\bar{b}\,WW^* \to b\bar{b}\,\ell\nu\,q\bar{q}'$, which has the same final state. The study has four parts:

1. Monte Carlo samples of $t\bar{t}$ and hh at parton level (MadGraph5), particle level (Pythia8) and detector level (Delphes, ATLAS card, no pileup), with the full $t\bar{t}$ reconstruction, and the ten angles computed at each level (Sections 2–3).

2. The density $f_{t\bar{t}}$ by OSDE in the paper's basis, its validation, the number of coefficients that matter, the orders needed at particle and detector level, and a comparison with a spherical-harmonic basis (Sections 4–7).

3. How well $f_{t\bar{t}}$ separates hh from $t\bar{t}$ compared with other unsupervised methods and with supervised machine learning on the same inputs, and why the separation seen at parton level disappears after reconstruction (Sections 9–10).

4. The fix: putting the momenta back through the resonance masses, which recovers the parton-level performance at particle and detector level, and the effect of realistic quark tagging (Section 11).

These four parts are Study A. Study B (Section 13) turns to new physics in the production with on-shell tops, a heavy scalar $H \to t\bar{t}$ at 500 GeV and at the $2m_t$ threshold, where the masses carry nothing and the angles everything, and introduces the OSDE coefficients as a bump hunt in angle space. Study C (Section 14) takes the method towards the detector: pileup, and a lookup-table version for the Level-1 trigger. The headline answers are collected in Table 1. Throughout, AUC is the area under the ROC curve for hh versus $t\bar{t}$ (0.5 = no separation), always evaluated on events that were not used in any fit: the independent $t\bar{t}$ sample, or the fit sample scored leave-one-out where stated.

The slides (`slides/slides.pdf`) show the same material in the same order; the section titles here match the slide sections. All numbers in this note are read from the analysis outputs by `note/make_note.py`, so the note and the slides cannot drift apart.

Table 1: Headline answers.

question	answer
Does $f_{t\bar{t}}$ separate hh from $t\bar{t}$ at parton level?	Yes: AUC 0.665 (0.685 with the lepton trigger), using no hh information
Is it the best unsupervised method?	Narrowly: $+0.011 \pm 0.003$ over a normalizing flow; well ahead of a Gaussian mixture and an autoencoder
How far from the best possible with these angles?	A supervised classifier on the same angles reaches 0.89
Does y help a supervised classifier?	No (change within ± 0.002)
What happens at gen and reco level?	AUC drops to 0.53 and 0.51
Why?	The χ^2 fit and the m_W-constrained neutrino rebuild hh events as $t\bar{t}$; the information survives in observables that do not assume $t\bar{t}$ (BDT AUC 0.87)
How many coefficients matter?	Cross-validation keeps 326, 385, 459, 342 of 14 400 (parton, parton+trig, gen, reco); $\sim$20 terms do as well as all of them
Can the momenta be put back?	Yes: one Breit–Wigner factor per resonance mass. Parton 0.65$\to$1.00, gen 0.53$\to$0.69, reco 0.51$\to$0.60

Part I

Study A: $t\bar{t}$ versus $hh \to b\bar{b}\,WW^*$

2 Samples and event levels

2.1 Generation

All samples are produced by `generate_samples.py`: MadGraph5_aMC@NLO 3.6.7 at leading order (4-flavour proton, $\sqrt{s} = 13$ TeV), Pythia8 for the parton shower and hadronization, and Delphes 3.5.1 with the ATLAS card and no pileup. The $t\bar{t}$ process is generated with the full decay chain so that spin correlations are kept and without generator-level cuts: $pp \to t\bar{t}$, $t \to bW^+(\to \ell^+\nu)$, $\bar{t} \to \bar{b}W^-(\to q\bar{q}')$ plus the charge conjugate, with $\ell = e, \mu$. The cross section is $\sigma = 136.5$ pb, i.e. the LO $t\bar{t}$ cross section times $2\,\mathcal{B}(W \to \ell\nu)\,\mathcal{B}(W \to q\bar{q}')$. Two statistically independent $t\bar{t}$ samples are used throughout: the *fit sample* (100k events, seed 1001), on which every density is fitted, and the *independent sample* (10k events, seed 12345), on which every result is evaluated.

The hh sample (20k events) uses MadGraph's built-in `heft` model (effective ggh and $gghh$ couplings in the $m_t \to \infty$ limit): $gg \to hh$, $h \to b\bar{b}$, $h \to \ell\nu q\bar{q}'$ generated as one process. Because $h \to WW^*$ has one off-shell W, the $h \to \ell\nu q\bar{q}'$ decay is a four-body decay. Its cross section times branching ratio, 0.112 fb, is the LO HEFT value with MadGraph defaults and not an SM prediction, so all hh distributions are shown at unit area. For $t\bar{t}$-hypothesis quantities hh goes through exactly the same reconstruction; at parton level the pairing of the b quarks with the W bosons is the one closest to the $t\bar{t}$ hypothesis (both m_{Wb} closest to m_t).

2.2 Event levels and selection

Four event levels are used (Table 2). *Parton* is the MadGraph output with no cuts. *Parton+trig* adds only the offline single-lepton (trigger) cuts to the charged lepton, $p_T > 25$ GeV and $|\eta| < 2.47$ (e) or 2.5 (μ), and keeps 71.7% of the events. *Gen* is the particle level: anti-k_t $R = 0.4$ GenJets,

the prompt lepton, and the missing momentum from the sum of all neutrinos. Prompt leptons are status-1 e/μ from a W, Z or h, or from a τ that itself comes from one of these (in $h \to WW^*$ the lepton can come directly from the h in the record); leptons from B and D decays, also via τ, are not prompt. GenJets within $\Delta R < 0.4$ of a prompt lepton are removed, since the lepton is clustered into a jet otherwise. A GenJet is b-labelled if a b quark with $p_T > 5$ GeV lies within $\Delta R < 0.4$. *Reco* is the Delphes output: jets, electrons, muons and the calorimeter missing transverse momentum, with the Delphes b-tag bit. Jets need $p_T > 25$ GeV and $|\eta| < 2.5$; leptons the same cuts as the trigger.

The gen and reco selection requires exactly one lepton, at least four jets and at least one b-jet, followed by the χ^2 $t\bar{t}$ reconstruction

$$\chi^2 = \left(\frac{m_{jj} - m_W}{\sigma_W}\right)^2 + \left(\frac{m_{jjb} - m_t}{\sigma_t}\right)^2 + \left(\frac{m_{\ell\nu b} - m_t}{\sigma_t}\right)^2, \tag{1}$$

minimized over all assignments of the six leading jets to $(b_{\mathrm{had}}, b_{\mathrm{lep}}, q_1, q_2)$ with $\min(n_b, 2)$ b-tagged jets in the two b slots, and over the two solutions of the neutrino p_z from the m_W constraint, so that the m_t term resolves the two-fold ambiguity ($\sigma_W = 10$, $\sigma_t = 15$ GeV). The cutflow is in Table 3: 41.3% of the $t\bar{t}$ events survive at gen level and 24.4% at reco level; hh leptons are softer (from the W^*) and less isolated, so its efficiency is lower. Reco and gen missing momentum agree for muon and electron events (reco minus gen along the lepton direction: -5.9 vs -5.2 GeV), which checks that the calorimeter missing momentum does not lose the muons.

Table 2: Event levels.

	parton	parton+trig	gen (particle)	reco
source	MG5 LHE	MG5 LHE	Pythia8	Delphes ATLAS
selection	none	ℓ cuts only	$1\ell, \geq 4j, \geq 1b, \chi^2$	$1\ell, \geq 4j, \geq 1b, \chi^2$
events kept	100%	71.7%	41.3%	24.4%
E_T^{miss}	p_T^{ν}	p_T^{ν}	$\sum$ all ν	$-\sum$ calo towers

Table 3: Cutflow of the independent $t\bar{t}$ sample and of the hh sample (reco); fractions relative to all generated events.

selection	gen		reco		hh reco	
All events	10000	100.0%	10000	100.0%	20000	100.0%
exactly 1 lepton	7071	70.7%	4907	49.1%	5403	27.0%
≥ 4 jets	4196	42.0%	2830	28.3%	2133	10.7%
≥ 1 b-jet	4161	41.6%	2454	24.5%	1887	9.4%
≥ 2 b-jets	3305	33.1%	1221	12.2%	920	4.6%
χ^2 reconstruction (from ≥ 1 b)	4134	41.3%	2440	24.4%	1878	9.4%

2.3 Reconstruction quality

All four jets are assigned correctly in 23% (gen) and 18% (reco) of the reconstructed $t\bar{t}$ events. The reconstructed masses (Table 4) have broad inclusive peaks that come from wrong jet choices and from quarks outside the acceptance, not from the jet response: with the correct jets the hadronic top and W peaks are narrow.

Table 4: Mass resolution: median ± 68% half-width (GeV).

	$m_{t,\mathrm{had}}$, all	$m_{t,\mathrm{had}}$, correct jets	$m_{W,\mathrm{had}}$, all	$m_{W,\mathrm{had}}$, correct jets
gen	182 ± 47	175 ± 14	85 ± 29	83 ± 9
reco	177 ± 44	170 ± 16	84 ± 25	80 ± 10

The neutrino p_z (Figure 1) has a 68% half-width of 54 GeV at gen and 72 GeV at reco level, improving to 49 GeV when all jets are correct. The m_W constraint has no real solution in 32% of the events (the discriminant is then set to zero). When two real solutions exist, the top-mass term of the χ^2 picks the one closer to the truth in 63% of the events, which is not better than simply taking the smaller $|p_z|$ (65%).

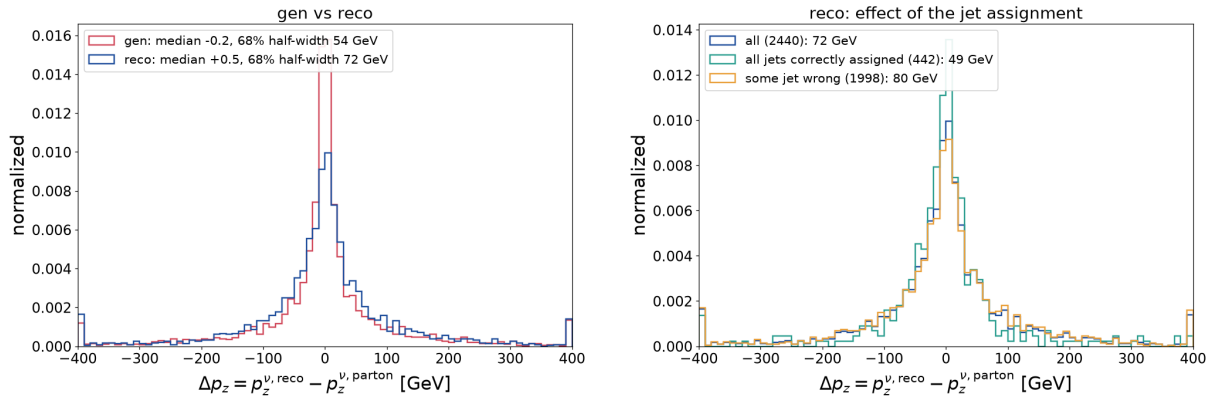


Figure 1: Neutrino $\Delta p_z = p_z^{\nu,\mathrm{reco}} - p_z^{\nu,\mathrm{parton}}$ at gen and reco level (left) and at reco level split by the correctness of the jet assignment (right).

For hh the reconstruction under the $t\bar{t}$ hypothesis is already visible in the masses: the hadronic boson is the off-shell W^* in 50% of the events, with a true median mass of 32 GeV, but the χ^2 picks jet pairs near m_W (median 95 GeV at gen and 89 GeV at reco level). This is the first sign of the effect studied in Section 10.

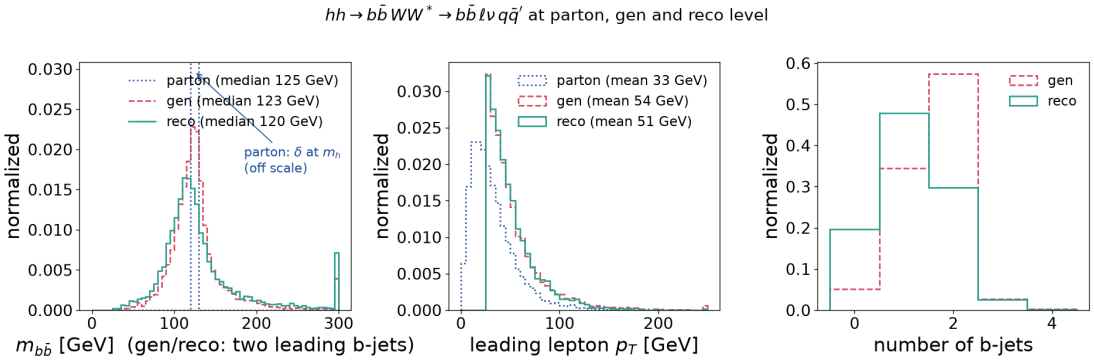
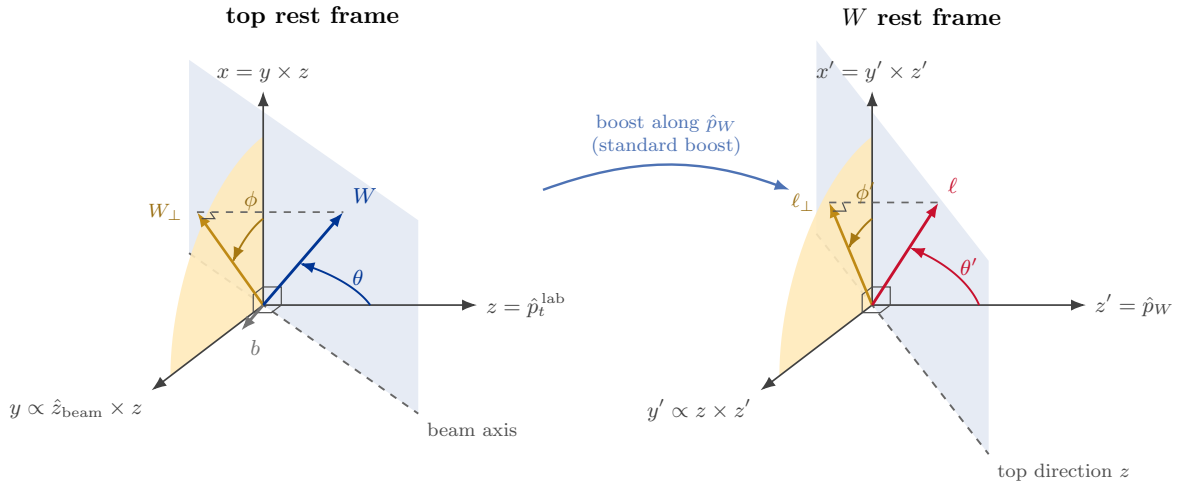


Figure 2: The hh sample at parton, gen and reco level: $m_{b\bar{b}}$, the hadronic W mass and the χ^2-reconstructed top mass.

3 The ten angles

The angles are defined exactly as in the paper (Figure 3). Y is the rapidity of the $t\bar{t}$ system in the laboratory and Θ the polar angle of the leptonic top in the $t\bar{t}$ rest frame. For each top, (θ, ϕ) are the polar and azimuthal angles of the W in the top rest frame, with $z = \hat{p}_t^{\text{lab}}$, $y \propto \hat{z}_{\text{beam}} \times z$ and $x = y \times z$; and (θ', ϕ') are the angles of the fermion in the W rest frame reached by a standard boost from the top frame, with $z' = \hat{p}_W$, $y' \propto z \times z'$ and $x' = y' \times z'$, i.e. the rotation $D(\phi, \theta, 0)$. On the leptonic side the fermion is the charged lepton; on the hadronic side (tilde angles) it is the down-type quark, which plays the same role in the $V-A$ decay. At gen and reco level the down-type jet is chosen by matching to the parton-level quarks, i.e. perfect up/down tagging as the paper assumes; realistic tagging is studied in Section 11.6. The implementation was checked by building momenta from known angles and recovering them, which agrees to 10^{-14}.



x–z plane (contains the beam and $\hat{p}_t^{\text{lab}}$; right: the x'–z' plane, contains $\hat{p}_W$ and the top direction) xy plane, perpendicular to z

θ: angle between z and the W. ϕ: **azimuth of the W around z**: project the W onto the xy plane ($W_\perp$); ϕ is the angle from x to $W_\perp$, toward y.
(θ', ϕ'): the same for the charged lepton in the W rest frame, with (x', y', z'): $z' = \hat{p}_W$, $y' = z \times z'/|z \times z'|$ (in the xy plane), $x' = y' \times z'$, i.e. $(x', y', z') = R(\phi, \theta, 0)\,(x, y, z)$.
The beam is not along z (z is the top direction); it lies in the x–z plane with a negative x component, since $y \propto \hat{z}_{\text{beam}} \times z$.
Hadronic top: same construction with the down-type quark in place of ℓ.

Figure 3: Definition of the angles in the top rest frame (left) and in the W rest frame (right); the tilde angles are defined in the same way for the hadronic top.

3.1 Validation

The W helicity fractions obtained from the moments of $\cos\theta'$ (Table 5) reproduce the SM expectation at parton level, $F_0 = m_t^2/(m_t^2 + 2m_W^2) \approx 0.70$, $F_L \approx 0.30$, $F_R \approx 0$, on both the leptonic and the hadronic side. At gen and reco level they are distorted by acceptance and combinatorics; no unfolding is attempted. At parton level the leptonic and hadronic decay angles have the same distributions, as they must in the SM (Kolmogorov–Smirnov p-values 0.84, 0.27, 0.24, 0.54 for $\cos\theta$, ϕ, $\cos\theta'$, ϕ'). At reco level they differ even when every jet is correctly assigned ($\cos\theta'$ KS distance 0.097), because the two sides are measured differently: lepton cuts and the neutrino solution on one side, jet cuts and jet response on the other.

Table 5: W helicity fractions from the moments of $\cos\theta'$.

	F_0	F_L	F_R
parton, ℓ	0.715 ± 0.013	0.288 ± 0.008	-0.003 ± 0.008
parton, down-type q	0.705 ± 0.013	0.298 ± 0.008	-0.003 ± 0.008
gen, ℓ	0.769 ± 0.019	0.107 ± 0.012	0.123 ± 0.012
gen, down-type q	0.954 ± 0.017	0.078 ± 0.011	-0.032 ± 0.011
reco, ℓ	0.687 ± 0.026	0.134 ± 0.017	0.179 ± 0.017
reco, down-type q	0.929 ± 0.023	0.086 ± 0.015	-0.014 ± 0.015

Table 6: Angle resolution with respect to parton level: correlation coefficient / median $|\Delta|$, for all reconstructed events and for events with all jets correctly assigned.

angle	gen, all	gen, correct	reco, all	reco, correct
Y	0.90 / 0.16	0.99 / 0.05	0.89 / 0.18	0.98 / 0.06
$\cos\Theta$	0.67 / 0.13	0.97 / 0.04	0.64 / 0.15	0.97 / 0.06
$\cos\theta$	0.70 / 0.14	0.89 / 0.06	0.62 / 0.22	0.84 / 0.11
ϕ	0.48 / 0.36	0.82 / 0.12	0.40 / 0.52	0.89 / 0.16
$\cos\theta'$	0.68 / 0.11	0.92 / 0.04	0.60 / 0.20	0.89 / 0.11
ϕ'	0.45 / 0.37	0.68 / 0.13	0.30 / 0.63	0.47 / 0.28
$\cos\tilde{\theta}$	0.59 / 0.18	0.98 / 0.04	0.45 / 0.24	0.97 / 0.05
$\tilde{\phi}$	0.40 / 0.46	0.88 / 0.09	0.34 / 0.69	0.90 / 0.09
$\cos\tilde{\theta}'$	0.41 / 0.18	0.97 / 0.04	0.31 / 0.25	0.96 / 0.06
$\tilde{\phi}'$	0.39 / 0.55	0.93 / 0.06	0.33 / 0.73	0.94 / 0.07

Reco angles track the parton angles when the jets are assigned correctly; the smearing of the inclusive distributions is dominated by combinatorics, not by the detector response.

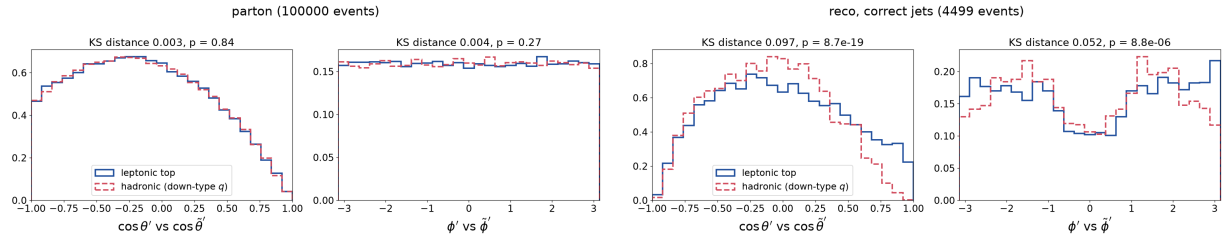


Figure 4: Leptonic versus hadronic W decay angles at parton level (left) and at reco level with correctly assigned jets (right).

4 The density $f_{t\bar{t}}$ by OSDE

4.1 Basis and estimator

The basis is the paper's: for each top the 24 functions

$$M^{j_1 j_2}_{m'm}(\phi,\theta,\phi',\theta') = \frac{\sqrt{2j_1+1}\sqrt{2j_2+1}}{4\pi} D^{j_1}_{m'm}(\phi,\theta,0)\, D^{j_2}_{m0}(\phi',\theta',0), \qquad j_1 \le 1,\ j_2 \le 2, \tag{2}$$

times normalized Legendre polynomials $p_l(\cos\Theta) = \sqrt{(2l+1)/2}\,P_l$ with $l < 5$ and normalized Hermite *functions* $h_n(Y) = (2^n n!\sqrt{\pi})^{-1/2} H_n(Y) e^{-Y^2/2}$ with $n < 5$; the bare Hermite polynomials would be orthogonal only with the weight e^{-Y^2}. The product $B = M_{\mathbf{j}} \tilde{M}_{\bar{\mathbf{j}}}\, p_l\, h_n$ has $24 \times 24 \times 5 \times 5 = 14400$ terms. The orthogonal-series estimate is $\hat{c} = \frac{1}{N}\sum_k B^*(\mathbf{x}_k)$ with its standard error, and $f_{t\bar{t}} = \sum \hat{c}\, B$, which is real because the set is closed under complex conjugation. The Wigner d functions are computed in the Jacobi-polynomial form, which is stable to $j = 24$.

Three separate densities are fitted: $f_{t\bar{t}}^{\text{parton}}$ on parton-level $t\bar{t}$ angles and applied to parton-level $t\bar{t}$ and hh, and likewise f^{gen} and f^{reco} (plus $f^{\text{parton+trig}}$). Each is fitted on the fit sample and evaluated on the independent sample and on hh. The code was checked with unit tests (`tests/test_osde.py`, 14 tests): the Wigner $d^j_{m'm}$ agree with closed forms and with `sympy` to 10^{-12}; the Gram matrices of M, p_l and h_n equal the identity to 10^{-12}; a toy density with known coefficients is recovered from 2×10^5 events with unit-width pulls; the fitted f has an imaginary part below 10^{-10} of the real part; the leave-one-out formula below equals brute-force refits; and the extended basis of Section 6 reduces exactly to the paper's basis.

4.2 Self-inclusion bias and leave-one-out

Each fit event adds $\frac{1}{N}\sum_j |B_j(\mathbf{x}_k)|^2$ to its own value of f, because its own contribution to every coefficient is evaluated at its own position. With 14 400 terms this self term is not negligible (Table 7): the fit sample looked systematically more $t\bar{t}$-like than the independent sample until the self term was removed. Removing it exactly (leave-one-out, LOO) makes the fit sample agree with the independent sample in the fraction of negative f, in the mean surprisal and in the Kolmogorov–Smirnov distance; the difference was entirely this bias, and largest at reco level where N is smallest. All plots use LOO values for the fit sample.

Table 7: Fit sample versus independent sample: the self-inclusion bias and its removal by leave-one-out (LOO).

level	self term / f	$f \le 0$: fit / fit LOO / indep.	$\langle s \rangle$: fit / fit LOO / indep.	KS vs indep.: fit / LOO
parton	7%	0.6% / 1.5% / 1.6%	11.98 / 12.08 / 12.07	0.041 / 0.010
parton+trig	8%	0.7% / 2.0% / 1.6%	11.81 / 11.91 / 11.90	0.048 / 0.008
gen	10%	1.0% / 2.7% / 2.7%	11.44 / 11.57 / 11.59	0.071 / 0.012
reco	16%	1.2% / 4.5% / 4.5%	11.33 / 11.53 / 11.53	0.103 / 0.012

4.3 Results

Table 8 summarizes the fits; Figures 5–10 show the one-dimensional projections of $f_{t\bar{t}}$ against the data and the distributions of $y = f_{t\bar{t}}$ for $t\bar{t}$ and hh. The negative values of f are the Gibbs overshoot of a truncated series, as noted in the paper; they are rare for $t\bar{t}$ and more frequent for hh, which lives where $f_{t\bar{t}}$ is small. The surprisal $s' = -\ln(y - y_0)$ with $y_0 = 2.5 \min y$ makes every event finite, and the ROC curve is obtained by scanning a cut on y (events with $f < y_{\text{cut}}$ are called hh-like). At parton level $f_{t\bar{t}}$ separates hh from $t\bar{t}$ with AUC 0.653 (0.669 after the lepton trigger); at gen and reco level the separation almost vanishes (0.525 and 0.507). Understanding and curing this loss is the subject of Sections 10 and 11.

Table 8: Summary of the $f_{t\bar{t}}$ fits in the paper's basis. $t\bar{t}$ numbers are from the independent sample; AUC is the probability that an hh event has lower $f_{t\bar{t}}$ than a $t\bar{t}$ event.

level	N_{fit}	N_{indep}	N_{hh}	$\lvert\hat{c}\rvert > 3\sigma$	$\int f$	$f \leq 0$: $t\bar{t}$ / hh	$\langle s \rangle$: $t\bar{t}$ / hh	AUC
parton	100000	10000	20000	186	1.006	1.6% / 5.9%	12.07 / 12.56	0.653
parton+trig	71322	7170	10149	255	1.000	1.6% / 6.8%	11.90 / 12.45	0.669
gen	41317	4134	4416	330	0.985	2.7% / 4.3%	11.59 / 11.66	0.525
reco	24030	2440	1878	253	0.987	4.5% / 4.8%	11.53 / 11.56	0.507

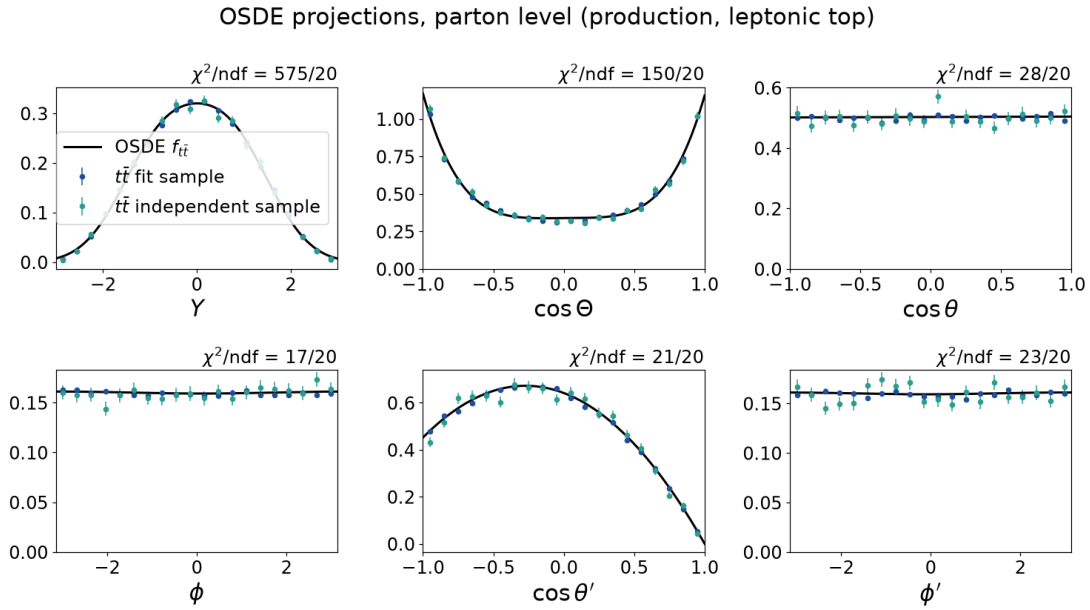


Figure 5: $f_{t\bar{t}}^{\text{parton}}$: one-dimensional projections (black) against the fit sample, production angles and leptonic top.

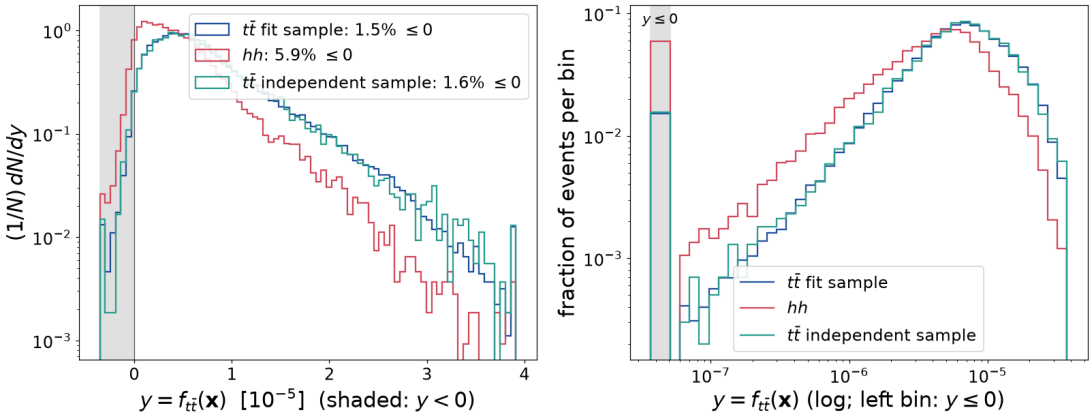


Figure 6: $y = f_{t\bar{t}}^{\text{parton}}$ for $t\bar{t}$ (independent sample and fit sample with LOO) and hh, linear (left, shaded: $y < 0$) and logarithmic (right; the left-most bin holds all events with $y \leq 0$).

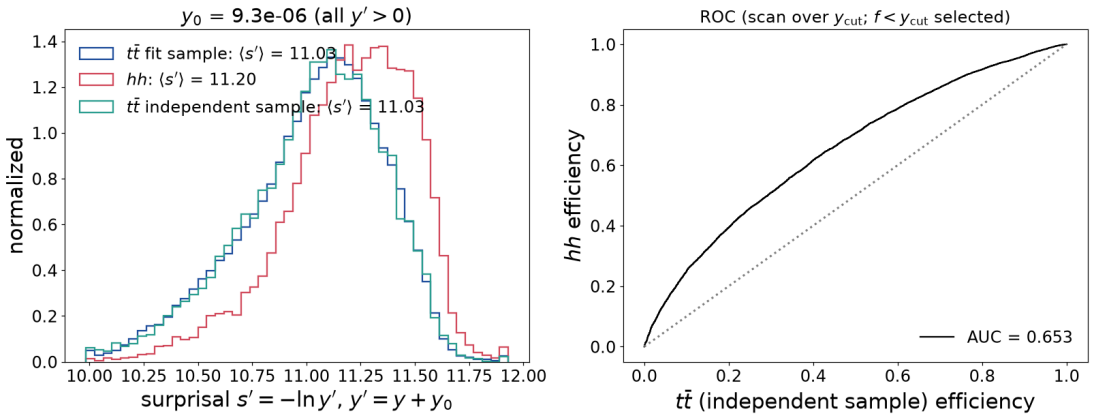


Figure 7: parton level: surprisal $s' = -\ln(y - y_0)$ (left) and the ROC curve from a scan over y_{cut} (right).

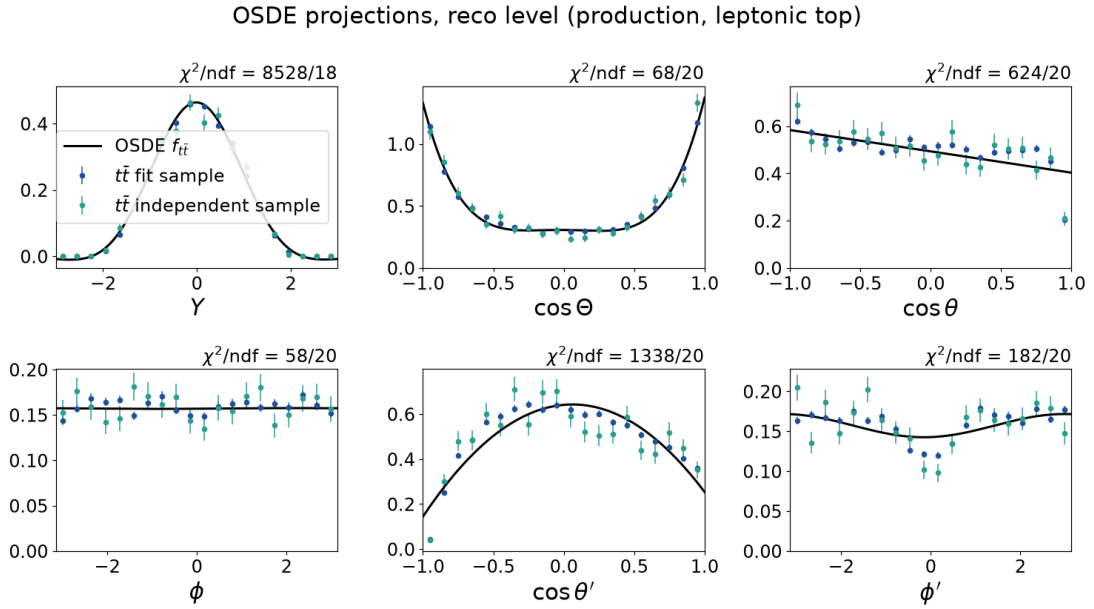


Figure 8: $f_{t\bar{t}}^{\text{reco}}$: one-dimensional projections (black) against the fit sample, production angles and leptonic top.

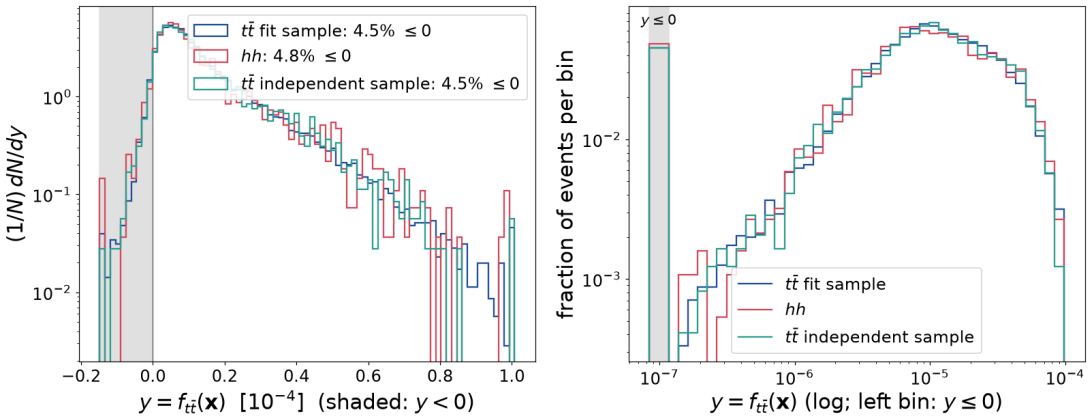


Figure 9: $y = f_{t\bar{t}}^{\text{reco}}$ for $t\bar{t}$ (independent sample and fit sample with LOO) and hh, linear (left, shaded: $y < 0$) and logarithmic (right; the left-most bin holds all events with $y \leq 0$).

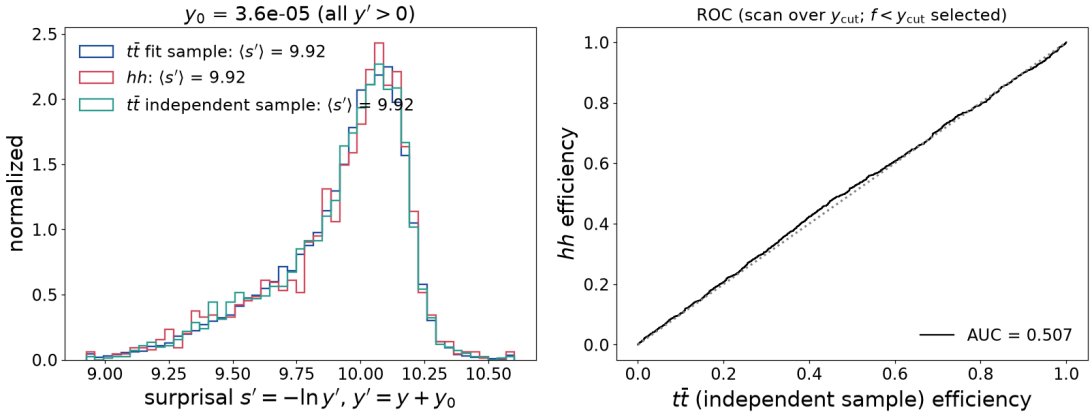


Figure 10: reco level: surprisal $s' = -\ln(y - y_0)$ (left) and the ROC curve from a scan over y_{cut} (right).

4.4 The Hermite scale

The width of the Hermite functions is a free parameter of the fit; the paper uses $s = 1$ and so does this note by default. The Y distribution is narrower at gen and reco level (std ≈ 0.8), and $s = 1$ then fits poorly: f overshoots below zero at large $|Y|$ (Table 9). The extended basis of Section 6 uses $s = \mathrm{std}(Y)$.

Table 9: χ^2/ndf of the Y projection against the fit sample for different Hermite scales s.

level	$s = 1.0$	$s = 0.9$	$s = 0.8$	$s = 0.7$	$s = \mathrm{std}(Y)$
parton	660/24	998/24	379/24	507/24	230/24
parton+trig	172/24	904/24	1013/24	117/24	111/24
gen	2298/20	2956/20	548/20	728/20	548/20
reco	4485/21	3864/21	721/21	361/21	515/21

4.5 The discriminator t

The paper's variable t is a monotonic rescaling of the surprisal: $y' = y - y_0$, $s' = -\ln y'$ and $t = (s'_1 - s')/(s'_1 - s'_0) \in [0, 1]$, so that $t \to 0$ is less $t\bar{t}$-like. Two conventions had to be inferred from the paper's figure. First y_0: its axis starts at about $2.5 \times \min y$ and its t distribution is populated from 0, which needs y_0 well below $\min y$; $y_0 = 2.5 \min y$ is used here. The shape of t depends on this choice (Figure 12), the ROC curve and AUC do not, since the map is monotonic. Second, the endpoints s'_0 and s'_1 are taken as the 0.5% and 99.5% quantiles of s' over $t\bar{t}$ and hh together, with the events beyond put into the first and last bins, so that t fills $[0, 1]$ with no empty bins. Figure 11 shows y and t in the style of the paper at parton level; the same figures for the other levels are in `osde/`.

5 How many coefficients matter?

5.1 Criterion

A term is kept if and only if its coefficient is significant, $|\hat{c}|/\sigma(\hat{c}) > t$, with the constant always kept. Conjugate partners have equal $|\hat{c}|$ and σ, so they are kept or dropped together and f stays real (checked). The Kronmal–Tarter rule is $t = \sqrt{2}$: dropping a term costs $|c|^2$ in integrated squared error, keeping it costs its variance σ^2, so a term is kept if $|\hat{c}|^2 > 2\sigma^2$. With 14 400 mostly null terms this still admits about 16% of the pure-noise terms, so t is instead chosen by five-fold cross-validation. For an orthonormal basis the cross-validated error needs no integrals: $\mathrm{ISE} - \int f^2 = \sum_{\text{kept}} |\hat{c}|^2 - 2\,\mathrm{Re} \sum_{\text{kept}} \hat{c}\, \hat{c}^*_{\text{held-out}}$. The minimum is at $t = 2.5$ and keeps 326, 385, 459, 342 of the 14 400 terms at parton, parton+trig, gen and reco level (Figure 13); the curve is flat out to 5σ (54, 108, 153, 127 terms).

5.2 What the significant terms are

Table 11 lists the leading terms at parton level. They are the production angle (P_2, P_4 of $\cos\Theta$, i.e. the forward–backward peaking of t-channel $gg \to t\bar{t}$), the width of the rapidity distribution (h_2, h_4 of Y), and the W helicity on each side ($j_1 = 0$, $j_2 = 1, 2$: functions of $\cos\theta'$ alone, i.e. the helicity fractions F_0, F_L, F_R), followed by their products. Table 12 classifies all terms above

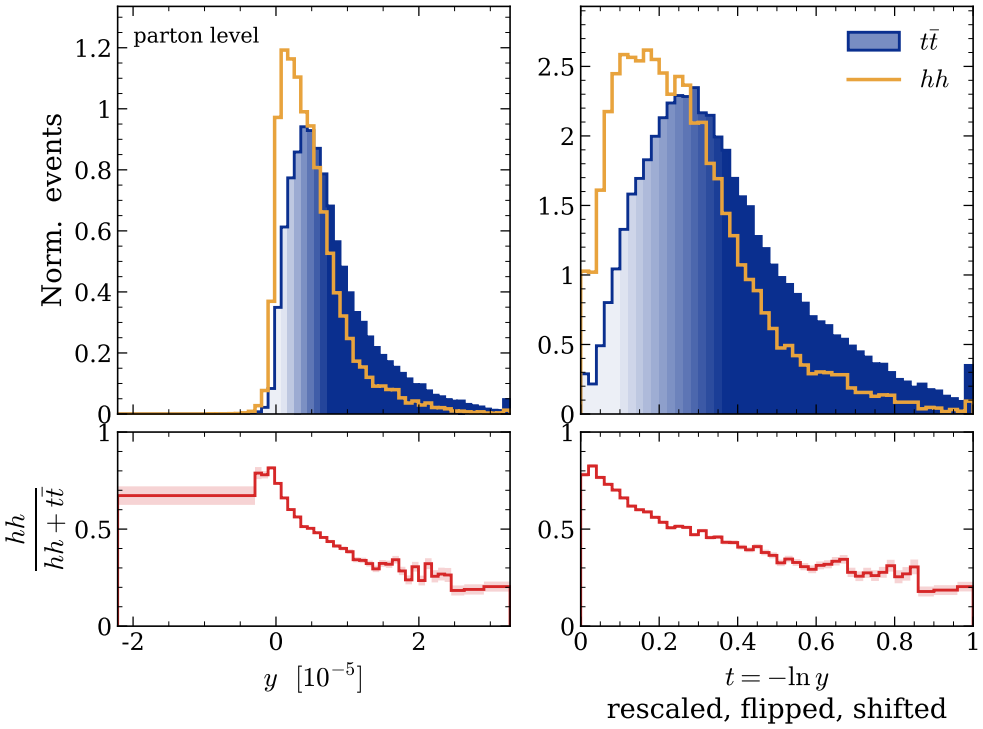


Figure 11: $y = f_{t\bar{t}}^{\text{parton}}$ and t for $t\bar{t}$ (fit sample, LOO) and hh at unit area, with the ratio $hh/(hh + t\bar{t})$ below, drawn as in the paper.

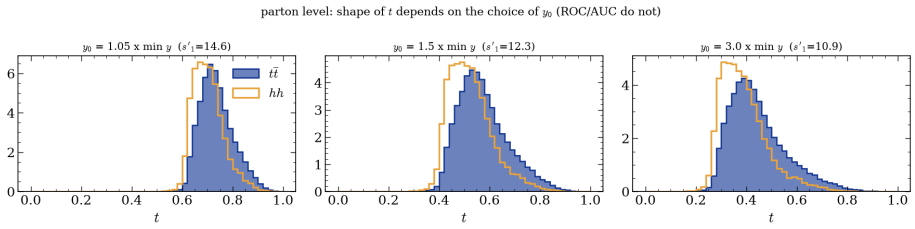


Figure 12: Shape of t for different choices of y_0 at parton level.

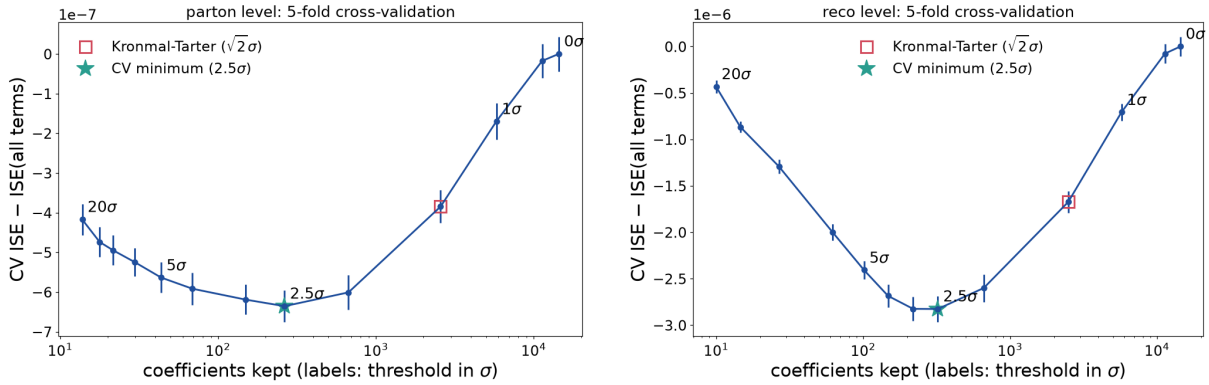


Figure 13: Cross-validated error versus the number of coefficients kept, parton (left) and reco level (right).

Table 10: Performance of the truncated $f_{t\bar{t}}$. The simplified density has fewer negative values and the same or better hh separation with 1–3% of the terms.

level	terms kept	number	AUC	$f \leq 0$ ($t\bar{t}$ indep.)	$f \leq 0$ (hh)
parton	all (paper)	14400	0.653	1.6%	5.9%
	$\sqrt{2}\sigma$ (K–T)	2629	0.658	1.0%	4.8%
	CV best	326	0.665	0.2%	2.4%
	5σ	54	0.669	0.1%	2.1%
parton+trig	all (paper)	14400	0.669	1.6%	6.8%
	$\sqrt{2}\sigma$ (K–T)	2756	0.677	1.2%	5.7%
	CV best	385	0.685	0.4%	3.4%
	5σ	108	0.691	0.4%	3.7%
gen	all (paper)	14400	0.525	2.7%	4.3%
	$\sqrt{2}\sigma$ (K–T)	2725	0.526	1.9%	2.9%
	CV best	459	0.525	0.9%	1.3%
	5σ	153	0.527	1.0%	1.5%
reco	all (paper)	14400	0.507	4.5%	4.8%
	$\sqrt{2}\sigma$ (K–T)	2551	0.508	2.7%	2.8%
	CV best	342	0.505	1.1%	1.5%
	5σ	127	0.507	1.5%	1.9%

5σ by their physics content: at parton level the $t\bar{t}$ spin correlations are resolved, with imaginary parts consistent with zero (maximum 1.4σ) as expected without CP violation; at reco level the spin correlations are no longer resolved and acceptance-induced "polarization" terms appear instead.

Table 11: Leading terms at parton level, after the constant. M: leptonic top; $\tilde{M}$: hadronic top; $j_1 = 0$: W helicity; $j_1 = 1$: top spin.

#	term	$\lvert\hat{c}\rvert/\sigma$	physics
1	$P_2(\cos\Theta)$	115	production angle
2	$M^{01}_{0,0}$	88	W helicity (lep)
3	$\tilde{M}^{01}_{0,0}$	87	W helicity (had)
4	$M^{02}_{0,0}$	80	W helicity (lep)
5	$\tilde{M}^{02}_{0,0}$	79	W helicity (had)
6	$h_2(Y)$	59	rapidity
7	$P_4(\cos\Theta)$	34	production angle
8	$M^{01}_{0,0} \times P_2(\cos\Theta)$	32	W helicity (lep); production angle
9	$\tilde{M}^{01}_{0,0} \times P_2(\cos\Theta)$	31	W helicity (had); production angle
10	$M^{02}_{0,0} \times P_2(\cos\Theta)$	29	W helicity (lep); production angle

5.3 Why fewer terms do better: estimation noise

A density with about 20 terms separates hh from $t\bar{t}$ *better* than the full 14 400-term expansion (0.670 against 0.653 at parton level). This is not a fluctuation but estimation noise. Each $\hat{c}_k$ is a sample mean over N events and carries noise $\sigma_k \propto 1/\sqrt{N}$; most of the true coefficients are essentially zero, so those terms contribute only noise. Summed over thousands of terms the noise is a random wobble on $\hat{f}(\mathbf{x})$, unrelated to $t\bar{t}$ versus hh, which scrambles the ranking of events. Dropping the insignificant terms removes the wobble at the cost of a small bias: the usual bias–variance trade-off. Two predictions follow, and both hold (Table 13, Figure 14): the gain of the

Table 12: Physics content of the terms above 5σ.

content of the term	parton	parton+trig	gen	reco
production angle + top pol. $\times$ W hel. interference	–	10	30	26
rapidity + top pol. $\times$ W hel. interference	–	15	24	12
top pol. $\times$ W hel. interference + $t\bar{t}$ spin correlation	16	12	–	–
production angle + rapidity + top polarization	–	1	12	12
W helicity + production angle	12	7	8	6
W helicity + production angle + rapidity	1	2	11	11
production angle + rapidity + top pol. $\times$ W hel. interference	–	7	11	11
W helicity + top pol. $\times$ W hel. interference	–	10	4	4
total	**53**	**107**	**152**	**126**

truncation shrinks as N grows, like $1/\sqrt{N}$, while the truncated f is converged already at 12.5k events; and two independent fits on disjoint halves of the data rank the events almost identically with the truncated f but not with the full one.

Table 13: Truncation gain versus the size of the fit sample, at parton level; the test sample is always the independent $t\bar{t}$ sample. Gain errors are ± 0.002–0.003 (paired bootstrap).

fit events N	AUC, all terms	AUC, $\lvert c\rvert/\sigma > 10$	gain	score correlation of two disjoint fits: all / truncated
12k	0.602	0.661 (10 terms)	+0.058	0.46 / 0.989
25k	0.627	0.666 (16 terms)	+0.039	0.65 / 0.998
50k	0.643	0.669 (19 terms)	+0.026	0.79 / 1.000
100k	0.653	0.670 (23 terms)	+0.018	–

6 Higher orders for gen and reco: the extended basis

The paper's basis has $j_1 \leq 1$ and $j_2 \leq 2$: it is at most linear in $\cos\theta$, at most quadratic in $\cos\theta'$, and contains only the first harmonic in ϕ'. This is complete for the physical decay distribution, and parton level confirms it, but the acceptance cuts create sharp edges ($\cos\theta \to 1$, $\cos\theta' \to -1$, $\phi' \approx 0$) that need higher orders. At gen level the one-dimensional projections of the paper-basis f miss the data badly (Figure 15); a direct one-dimensional expansion at the same order misses it in the same way (maximum difference 1.9%), while a higher-order expansion fits (χ^2 for 25 bins: 1787, 1251, 572, 1048, 1000, 657 at the paper's order against 32, 13, 17, 15, 15, 14). So the order, not the fit, limits f.

A one-dimensional cross-validated order scan per angle (Table 14) gives the orders each level needs. Parton level needs exactly the paper's orders apart from $\cos\Theta$ and Y; gen and reco need $j_{1,2}$ up to 13–24. A full product basis at those orders would have more than 10^6 terms per top pair, so an *extended block basis* is used instead: the union of disjoint subsets of the full orthonormal product basis, which is still orthonormal so that the OSDE, the leave-one-out and the cross-validated truncation are unchanged. Block A is the paper's full product at the physics order; blocks L and H are high-order single-top blocks (all (θ, ϕ) terms and all $m' = 0$ terms of one top with the other top constant); blocks P and Y are high-order $\cos\Theta$ and Y terms, with the Hermite scale set to the standard deviation of Y. The orders are capped at 16; the resulting basis has 14436 (parton), 18936 (parton+trig), 32494 (gen), 49867 (reco) terms. The price is that correlations between the two tops stay at the physics order.

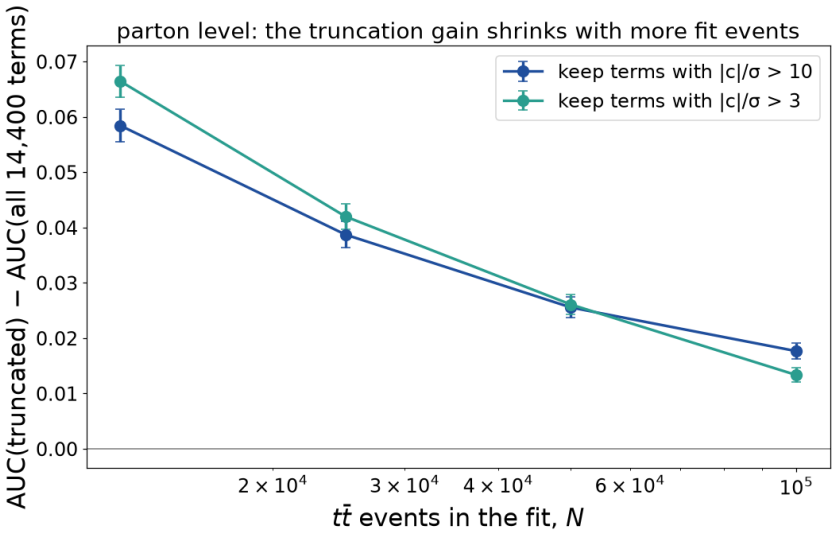


Figure 14: AUC gain of the truncated f over the full expansion versus the number of fit events.

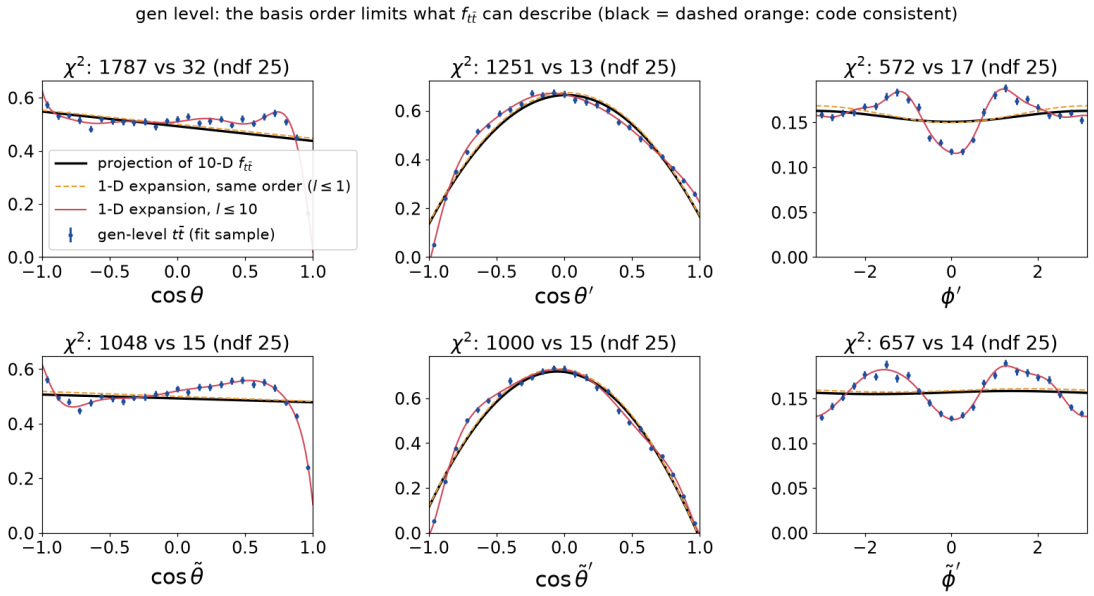


Figure 15: Gen level, paper basis: projections of f (same order) against a higher-order one-dimensional expansion.

Table 14: Order needed by each angle (one-dimensional cross-validated order: Legendre degree for cosines, harmonic for azimuths, Hermite order for Y) and the resulting requirement on j_1, j_2.

angle	parton	parton+trig	gen	reco
$\cos\theta$	0	4	13	18
ϕ	0	2	4	4
$\cos\theta'$	2	8	6	5
ϕ'	1	7	12	4
$\cos\tilde{\theta}$	1	1	7	7
$\tilde{\phi}$	0	0	4	3
$\cos\tilde{\theta}'$	2	2	8	24
$\tilde{\phi}'$	1	1	4	5
$\cos\Theta$	10	6	6	13
Y	10	10	8	24
needs	$j_1 \geq 1,\ j_2 \geq 2$	$j_1 \geq 4,\ j_2 \geq 8$	$j_1 \geq 13,\ j_2 \geq 12$	$j_1 \geq 18,\ j_2 \geq 24$

Table 15: χ^2/ndf of the one-dimensional projections against the $t\bar{t}$ fit sample, paper and extended basis.

level	basis	Y	$\cos\Theta$	$\tilde{\phi}'$	$\cos\tilde{\theta}'$	$\tilde{\phi}$	$\cos\tilde{\theta}$	ϕ'	$\cos\theta'$	ϕ	$\cos\theta$
parton	paper	28.8	7.5	1.7	0.6	1.4	1.4	1.1	1.1	0.8	1.4
	paper, CV	28.6	6.9	1.9	0.6	1.4	1.6	1.3	1.1	0.9	1.4
	extended	0.5	0.5	1.5	0.4	1.1	1.2	0.9	0.8	0.6	1.2
	extended, CV	0.5	0.8	1.7	0.4	1.2	1.3	1.1	0.8	0.7	1.2
parton+trig	paper	7.9	2.7	1.3	0.3	1.1	1.1	47.0	2.5	0.8	2.7
	paper, CV	7.8	2.7	1.4	0.3	1.2	1.1	45.7	2.5	0.9	2.7
	extended	0.8	0.6	0.9	0.2	0.4	0.8	0.6	0.7	0.3	0.7
	extended, CV	0.8	1.0	1.4	0.3	1.2	1.1	1.8	1.4	0.4	0.9
gen	paper	1057.0	4.5	32.4	37.5	8.0	44.7	29.7	40.9	5.3	68.3
	paper, CV	906.9	4.6	32.5	39.4	8.2	44.7	30.1	40.9	5.6	68.3
	extended	16.1	1.3	0.4	0.5	0.2	0.2	0.2	0.4	0.1	0.7
	extended, CV	43.0	1.3	1.0	0.7	1.0	0.6	1.4	0.4	0.9	1.3
reco	paper	473.8	3.4	22.1	24.6	2.1	25.8	9.1	66.9	2.9	31.2
	paper, CV	473.3	3.6	22.1	24.2	2.2	26.6	9.1	68.8	2.9	31.2
	extended	9.6	0.4	0.3	0.7	0.2	1.1	0.5	0.4	0.1	0.6
	extended, CV	1.6	0.9	1.6	2.4	1.1	1.7	1.5	1.3	0.8	1.8

After the cross-validated truncation the extended basis describes every angle at every level with χ^2/ndf between about 0.4 and 2 (Table 15, Figure 16), whereas the paper basis is good only at parton level. Yet this much better description of $t\bar{t}$ hardly changes the hh separation at gen and reco level (Table 16): the limitation is not the density but the $t\bar{t}$-hypothesis reconstruction (Section 10). Without truncation the many high-order terms add noise, with up to 13% negative f.

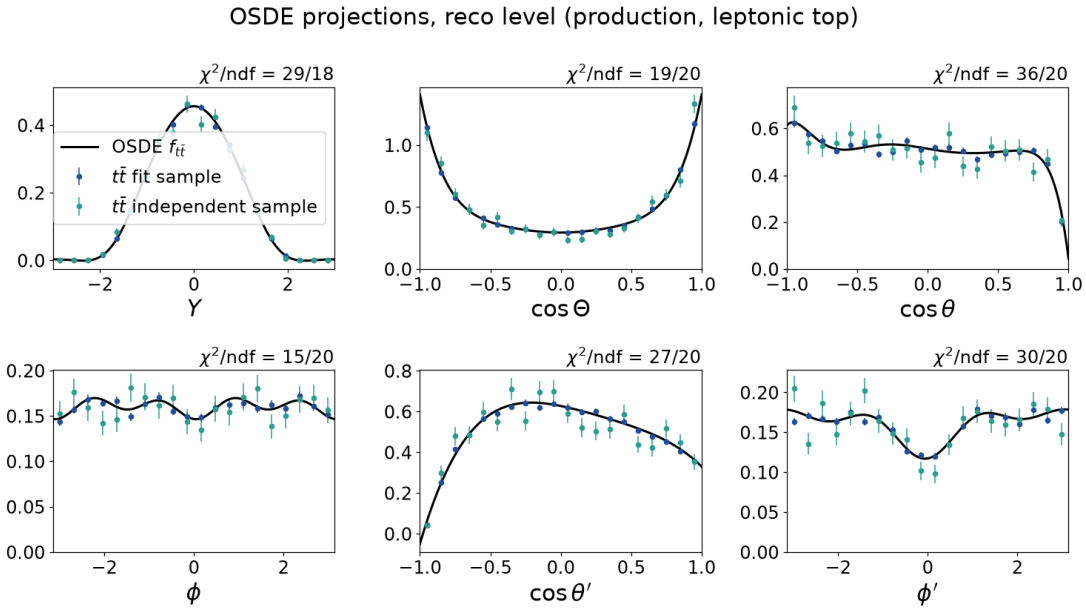


Figure 16: Extended basis at reco level after CV truncation: projections.

Table 16: Performance of the extended basis.

level	terms	number	AUC (hh vs $t\bar{t}$)	$f \leq 0$ ($t\bar{t}$ indep.)	$f \leq 0$ (hh)
parton	all terms	14436	0.655	1.4%	5.8%
	CV best	333	0.666	0.3%	2.5%
parton+trig	all terms	18936	0.667	2.1%	8.2%
	CV best	498	0.684	0.4%	4.3%
gen	all terms	32494	0.528	5.7%	8.4%
	CV best	679	0.535	1.6%	2.4%
reco	all terms	49867	0.507	12.7%	13.9%
	CV best	323	0.512	1.4%	1.8%

7 Spherical-harmonic basis

As an alternative to the M functions, products of spherical harmonics $Y_{l_1}^{m_1}(\theta,\phi)\,Y_{l_2}^{m_2}(\theta',\phi')$ with $l_1 \leq 1$, $l_2 \leq 2$ were used for each top: 36 functions instead of 24, 32 400 coefficients in all, with the same p_l and h_n. The set is orthonormal and closed under conjugation (tested), and the same OSDE, leave-one-out and cross-validated truncation apply. The physics difference is that M uses the spin-weighted $D^j_{m'm}(\phi,\theta,0)$: for example $d^1_{01} \propto \sin\theta$ is not a finite sum of Y_{l0}, so the SM decay density, which is finite in M, needs in principle infinitely many spherical-harmonic products. The

figure of merit is the held-out integrated squared error on the independent $t\bar{t}$ sample, $\mathrm{ISE} - \int f^2 = \sum |\hat{c}|^2 - 2\langle \hat{f} \rangle_{\mathrm{indep}}$, lower being closer to the true density; differences between bases are computed on the same events (paired), which makes their errors small.

With all terms kept, the spherical harmonics are worse than M at every level, by many standard deviations in ΔISE, simply because 32 400 terms are mostly noise. After the cross-validated truncation (Table 17, Figure 17) the two bases describe $t\bar{t}$ about equally well (ΔISE at parton level $+0.06 \pm 0.02$ in units of 10^{-6}, i.e. M is slightly better, about 3σ; at gen and reco level they are equal within errors), and the spherical harmonics need fewer terms (150 against 326 at parton level). The extended basis is the best at gen and reco level. The hh AUC hardly depends on the basis: once only the significant terms are kept, both bases describe the same few physical features.

Table 17: Comparison of the bases on the independent $t\bar{t}$ sample. $\Delta\mathrm{ISE} < 0$: closer to the true density than the paper's M basis (paired). Worst χ^2/ndf of the projections excludes Y.

level	basis	terms	ΔISE vs M $[10^{-6}]$	worst χ^2/ndf	$f \leq 0$	AUC
parton	paper (M), all	14400	–	7.5	1.6%	0.653
	paper (M), CV	326	–	6.9	0.2%	0.665
	sph. harmonics, all	32400	$+1.05 \pm 0.07$	7.5	4.0%	0.639
	sph. harmonics, CV	150	$+0.06 \pm 0.02$	7.3	0.1%	0.670
	extended, all	14436	-0.08 ± 0.01	1.5	1.4%	0.655
	extended, CV	333	-0.02 ± 0.01	1.7	0.3%	0.666
parton+trig	paper (M), all	14400	–	47.0	1.6%	0.669
	paper (M), CV	385	–	45.7	0.4%	0.685
	sph. harmonics, all	32400	$+1.40 \pm 0.11$	8.2	4.5%	0.648
	sph. harmonics, CV	269	$+0.07 \pm 0.04$	7.4	0.3%	0.688
	extended, all	18936	-0.04 ± 0.04	0.9	2.1%	0.667
	extended, CV	498	-0.21 ± 0.03	1.8	0.4%	0.684
gen	paper (M), all	14400	–	68.3	2.7%	0.525
	paper (M), CV	459	–	68.3	0.9%	0.525
	sph. harmonics, all	32400	$+2.35 \pm 0.23$	68.3	6.6%	0.520
	sph. harmonics, CV	555	-0.02 ± 0.11	68.3	1.3%	0.525
	extended, all	32494	$+0.85 \pm 0.15$	1.3	5.7%	0.528
	extended, CV	679	-1.11 ± 0.08	1.4	1.6%	0.535
reco	paper (M), all	14400	–	66.9	4.5%	0.507
	paper (M), CV	342	–	68.8	1.1%	0.505
	sph. harmonics, all	32400	$+3.77 \pm 0.43$	66.9	9.9%	0.507
	sph. harmonics, CV	279	-0.01 ± 0.14	64.5	1.5%	0.510
	extended, all	49867	$+6.34 \pm 0.32$	1.1	12.7%	0.507
	extended, CV	323	-0.92 ± 0.11	2.4	1.4%	0.512

8 Acceptance

To separate the effect of the selection from that of the reconstruction, Figure 18 shows the parton-level angles before any cut and after the gen and reco selections, so that the angle value of an event never changes and every shape difference is acceptance only; the angles measured at reco level are overlaid (dotted) to show what resolution and jet assignment add. The selection sculpts the angles in a few places. $\cos\theta' \to -1$ and $\phi' \approx 0$ are depleted because the lepton is emitted against the top's flight direction and fails the $p_T > 25$ GeV cut, already at gen level, so it is not an isolation effect. $|Y| \gtrsim 2$ is cut by the $|\eta| < 2.5$ acceptance. $\cos\theta$ and $\cos\tilde{\theta} \to +1$ are depleted because the

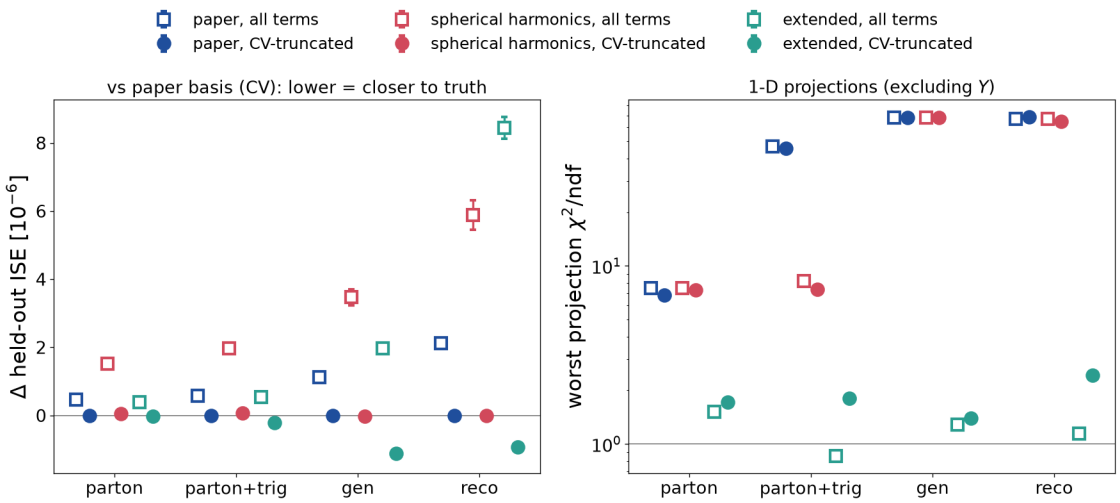


Figure 17: Paired held-out ISE difference with respect to the paper basis (left) and worst projection χ^2/ndf (right), per level and basis.

b quark then goes backward and is too soft for the jet cut. $\cos\Theta$ and the azimuthal angles have a nearly flat efficiency. The reco efficiency of $t\bar{t}$ after each cumulative cut is 1 lepton: 48.3%, $\geq$4 jets: 27.8%, $\geq$1 b-jet: 24.1%, χ^2 reco: 24.0%; for hh it is 27.0%, 10.7%, 9.4%, 9.4%. For hh the dotted curves show the effect that dominates everything else: the $t\bar{t}$-hypothesis χ^2 turns the flat parton-level $\cos\Theta$ into $t\bar{t}$-like peaks at ±1. The reconstruction, more than the acceptance, makes hh look like $t\bar{t}$.

9 Is the OSDE approach helping? Comparison with machine learning

9.1 Setup

Every method is evaluated on the same test events, the independent $t\bar{t}$ sample (10000 events) and a held-out 40% of hh (8000 events); training uses the $t\bar{t}$ fit sample and 50% of hh, with 10% of each held out for validation. Errors are bootstrap errors; differences between methods are paired bootstraps on the same test events. The *unsupervised* methods are trained on $t\bar{t}$ only, exactly like $f_{t\bar{t}}$, on the same ten angles: a Gaussian-mixture density (32 components), an autoencoder scored by its reconstruction error, and a normalizing flow (RealNVP, 8 coupling layers). The *supervised* methods see labelled $t\bar{t}$ and hh and therefore set an upper bound for the inputs: a boosted decision tree (BDT, scikit-learn histogram gradient boosting), a multi-layer perceptron (MLP) and an FT-Transformer on (a) the ten angles and (b) the angles plus the parton four-momenta, and a particle transformer with pairwise invariant-mass and ΔR attention biases on the momenta alone. To test whether y helps a classifier, the same classifiers are trained with $y = f_{t\bar{t}}$ and s' added to their inputs (leave-one-out y for the $t\bar{t}$ training events). Two controls must give AUC $\approx$ 0.5: shuffled labels, and $t\bar{t}$ (fit sample) against $t\bar{t}$ (independent sample).

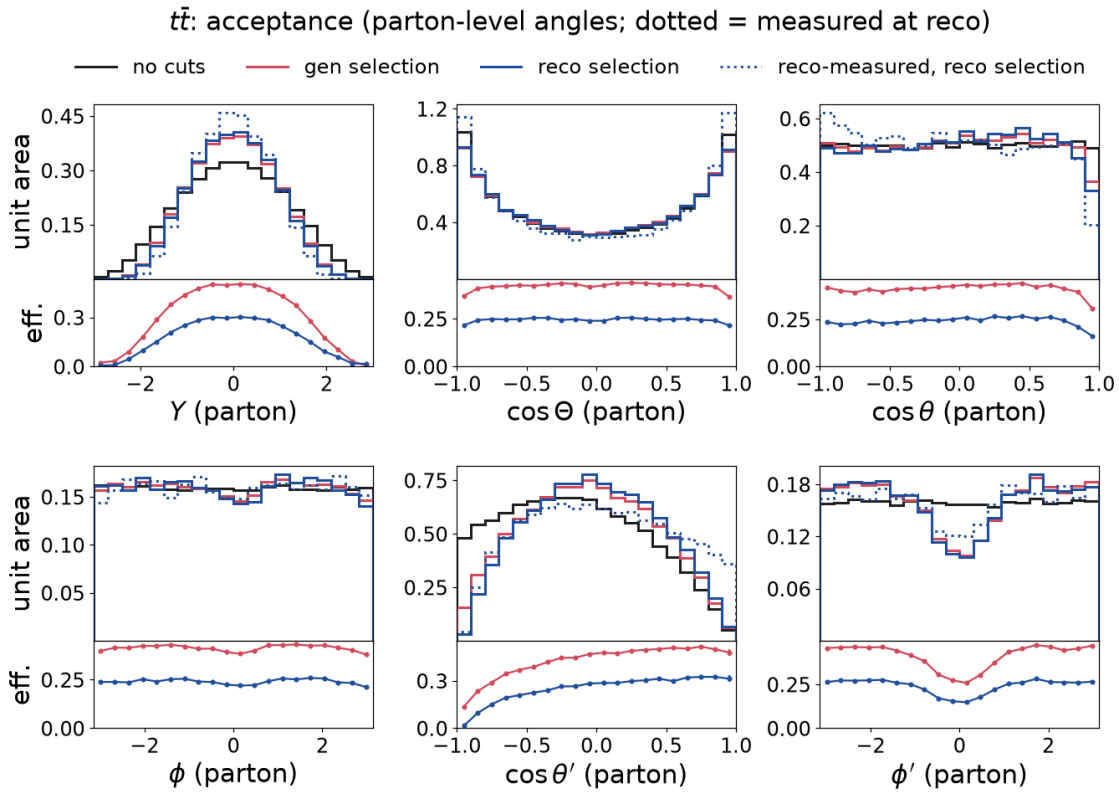


Figure 18: $t\bar{t}$: parton-level angles with no cuts, after the gen selection and after the reco selection (unit area), with the angles measured at reco level overlaid (dotted) and the efficiency below each panel.

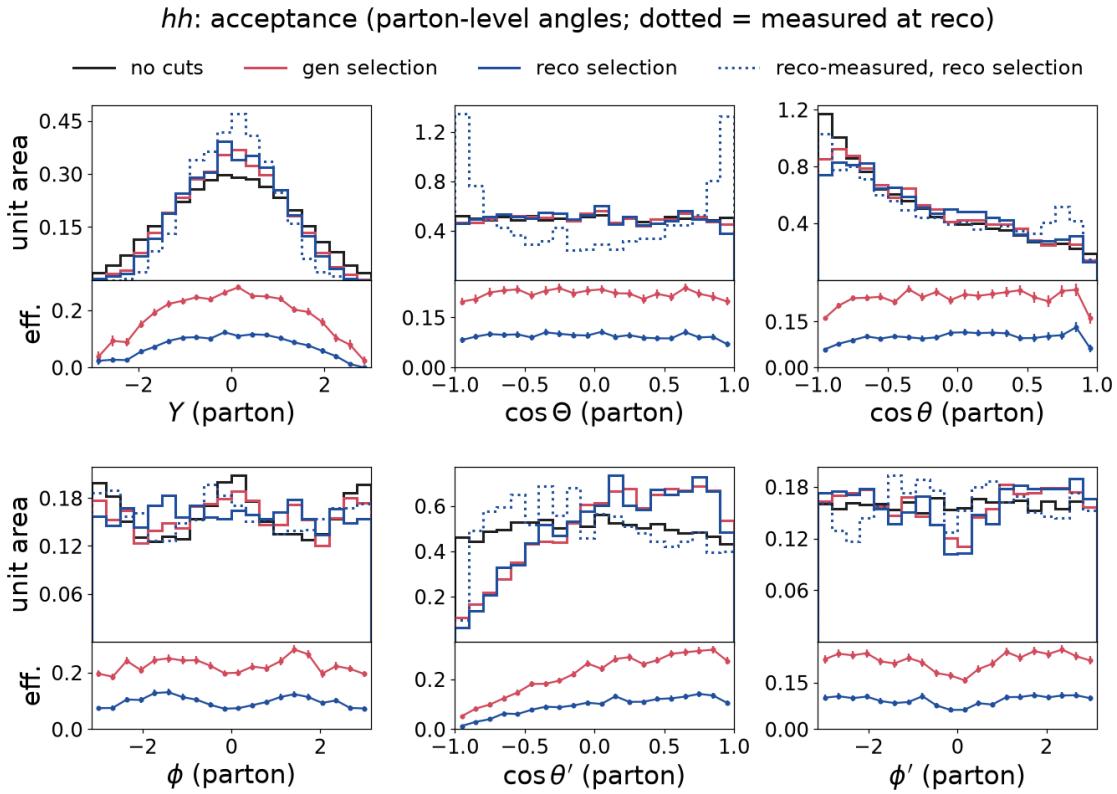


Figure 19: The same for hh.

9.2 The methods in plain words

The unsupervised methods all do the same job as $f_{t\bar{t}}$: learn the $t\bar{t}$ density from $t\bar{t}$ events and flag events where it is low. The *Gaussian mixture* models the density as a sum of 32 Gaussian blobs in the ten-dimensional angle space, with positions and widths fitted to $t\bar{t}$. The *autoencoder* is a network that squeezes the ten angles through a few numbers and rebuilds them; trained on $t\bar{t}$, it rebuilds unusual events badly, but it is not a density. The *normalizing flow* learns a smooth, invertible map $z = T(\mathbf{x})$ that turns the $t\bar{t}$ distribution into a standard Gaussian; the density then follows from the change of variables, $f(\mathbf{x}) = \mathcal{N}(T(\mathbf{x}))\,|\det \partial T/\partial \mathbf{x}|$. The mass map of Section 11, $u = \frac{1}{2} + \frac{1}{\pi}\arctan\frac{m-M}{W}$ with density $g_u\,du/dm$, is exactly a one-dimensional flow; RealNVP does the same in ten dimensions by stacking simple steps, each of which shifts and rescales half of the variables by amounts computed from the other half, so that the Jacobian stays cheap.

The supervised methods learn $P(hh\,|\,\mathbf{x})$ from labelled examples; the best any of them can do is the likelihood ratio $f_{hh}/f_{t\bar{t}}$, which $f_{t\bar{t}}$ alone can only approximate by acting as if f_{hh} were flat. The *BDT* is a few hundred small decision trees, each correcting the mistakes of the previous ones. The *MLP* is a standard neural network of weighted sums with simple non-linearities in between. The *FT-Transformer* turns each input variable into a vector (a token) and lets every variable attend to every other, so it learns which combinations matter. The *particle transformer* uses the six particles' four-momenta as tokens and is told each pair's invariant mass and ΔR, so it sees $m_{b\bar{b}} = m_h$ directly. They win because they know what hh looks like; the goal of $f_{t\bar{t}}$ is different, a $t\bar{t}$-only score that flags anything that is not $t\bar{t}$.

9.3 Results

Table 18: AUC (hh vs $t\bar{t}$) of every method on the same test events, parton level and parton+trig level.

inputs	method	AUC parton	AUC parton+trig
unsupervised	OSDE $f_{t\bar{t}}$, paper basis (all terms)	0.652 ± 0.004	0.667 ± 0.005
unsupervised	OSDE $f_{t\bar{t}}$, paper basis (CV-truncated)	0.665 ± 0.004	0.683 ± 0.005
unsupervised	Gaussian mixture density (32 comp.)	0.583 ± 0.005	0.589 ± 0.006
unsupervised	autoencoder (reconstruction error)	0.528 ± 0.004	0.588 ± 0.006
unsupervised	normalizing flow (RealNVP, 8 couplings)	0.654 ± 0.004	0.675 ± 0.005
angles	BDT	0.881 ± 0.002	0.882 ± 0.003
angles	MLP	0.887 ± 0.002	0.896 ± 0.003
angles	FT-Transformer	0.884 ± 0.002	0.886 ± 0.003
angles + y	BDT	0.879 ± 0.002	0.882 ± 0.003
angles + y	MLP	0.886 ± 0.002	0.895 ± 0.003
angles + y	FT-Transformer	0.886 ± 0.002	0.885 ± 0.003
angles + momenta	BDT	1.000 ± 0.000	0.998 ± 0.000
angles + momenta	MLP	1.000 ± 0.000	1.000 ± 0.000
angles + momenta + y	BDT	0.999 ± 0.000	0.998 ± 0.000
angles + momenta + y	MLP	1.000 ± 0.000	1.000 ± 0.000
momenta	Particle Transformer	1.000 ± 0.000	1.000 ± 0.000
control	control: BDT, shuffled labels	0.491 ± 0.005	0.517 ± 0.005
control	control: BDT, tt (fit) vs tt (independent)	0.502 ± 0.006	0.503 ± 0.007

As an anomaly detector that has never seen hh, the CV-truncated $f_{t\bar{t}}$ is the best of the unsupervised methods, narrowly ahead of the normalizing flow (paired difference $+0.011 \pm 0.003$ at parton level and $+0.008 \pm 0.003$ at parton+trig level) and well ahead of the Gaussian mixture and the autoencoder. A supervised classifier on the same ten angles reaches 0.89, the upper bound for these inputs. Adding y to the supervised classifiers changes their AUC by -0.002 to +0.002, within the uncertainty of ± 0.002: y carries no information that the classifier does not already learn from the angles. The momenta make the task trivial (AUC above 0.999), because they contain $m_{b\bar{b}} = m_h$ and the off-shell W^* mass; this is the scale of what the angles alone leave out, and the starting point of Section 11. The controls give 0.491 and 0.502 at parton level. At parton+trig level the shuffled-label control is 0.517 ± 0.006, about 3σ above 0.5; with some thirty AUCs computed, one such fluctuation is not surprising.

10 Why the separation washes out at gen level

10.1 Short answer

At parton level the hh "tops" are not tops: each pseudo-top is the W from one Higgs boson plus the b quark from the other, with a mass of median 136 GeV, and one of the two W bosons is far off shell (median 32 GeV); only 9% of the hh events have both pseudo-tops within 30 GeV of m_t. This shows in the angles: $\cos\theta$ and $\cos\tilde{\theta}$ peak at -1 for hh because the soft W^* goes backward while the b carries the pseudo-top's momentum, whereas they are flat for $t\bar{t}$; $\cos\Theta$ is isotropic for hh but forward–backward peaked for $t\bar{t}$ by t-channel top exchange; and $\cos\theta'$ has no $V-A$ zero at $+1$ for hh (Figure 22). At gen level the $t\bar{t}$-hypothesis reconstruction removes these differences. The χ^2 builds m_W and m_t from whatever jets fit best, picking the true four jets in only 8% of the hh events against 34% of $t\bar{t}$, and the m_W-constrained neutrino puts the off-shell leptonic W^* on

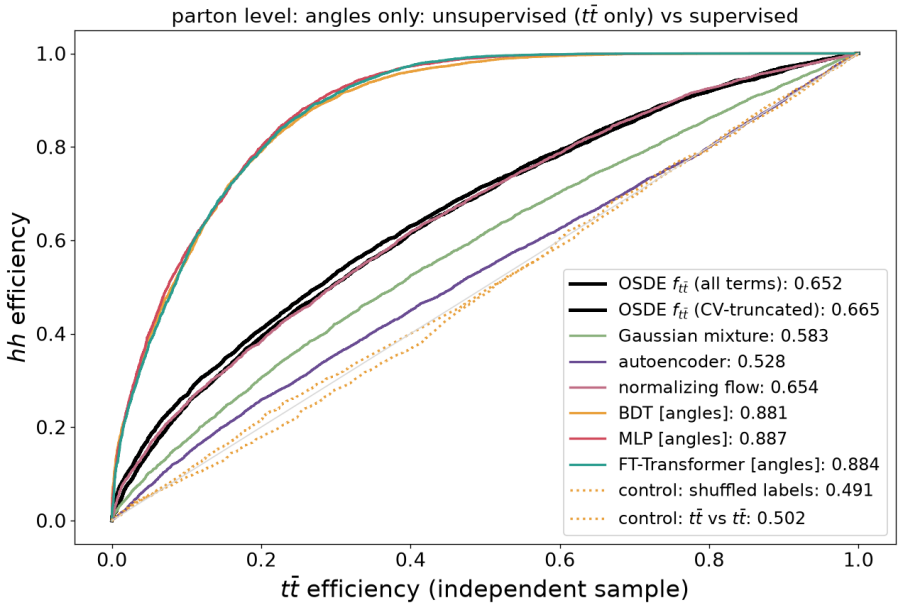


Figure 20: ROC curves at parton level, unsupervised and supervised methods on the ten angles.

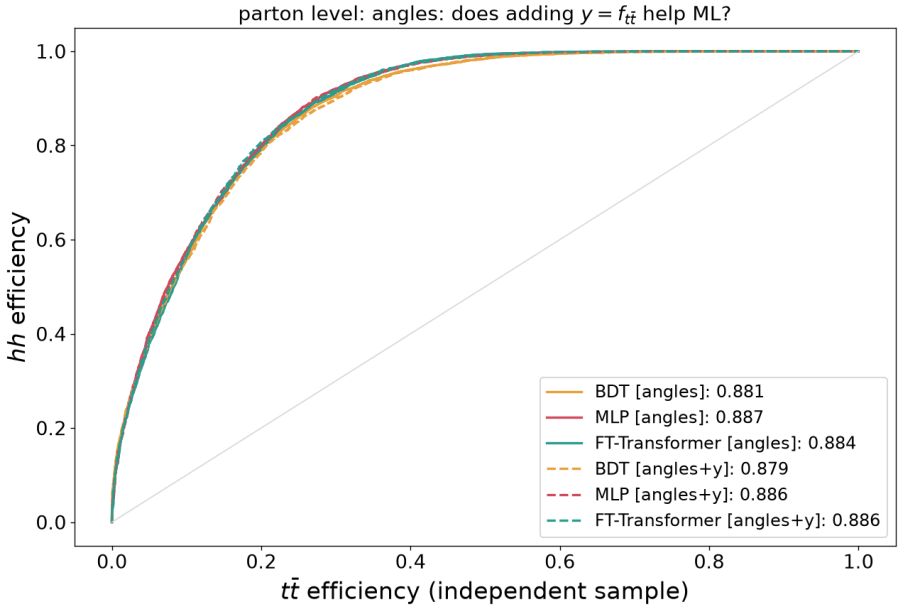


Figure 21: ROC curves at parton level: does adding y help a supervised classifier?

shell (its p_z is then off by a median 77 GeV, against 17 GeV for $t\bar{t}$). The information itself is still there at gen level: on the same events a BDT on observables that do not assume $t\bar{t}$ ($m_{b\bar{b}}$, ΔR_{bb}, E_T^{miss}, m_T, ...) reaches AUC 0.87, with $m_{b\bar{b}}$ alone at about 0.76 (likelihood-ratio AUC); the χ^2 value itself also differs (median 21.6 for hh against 6.2 for $t\bar{t}$).

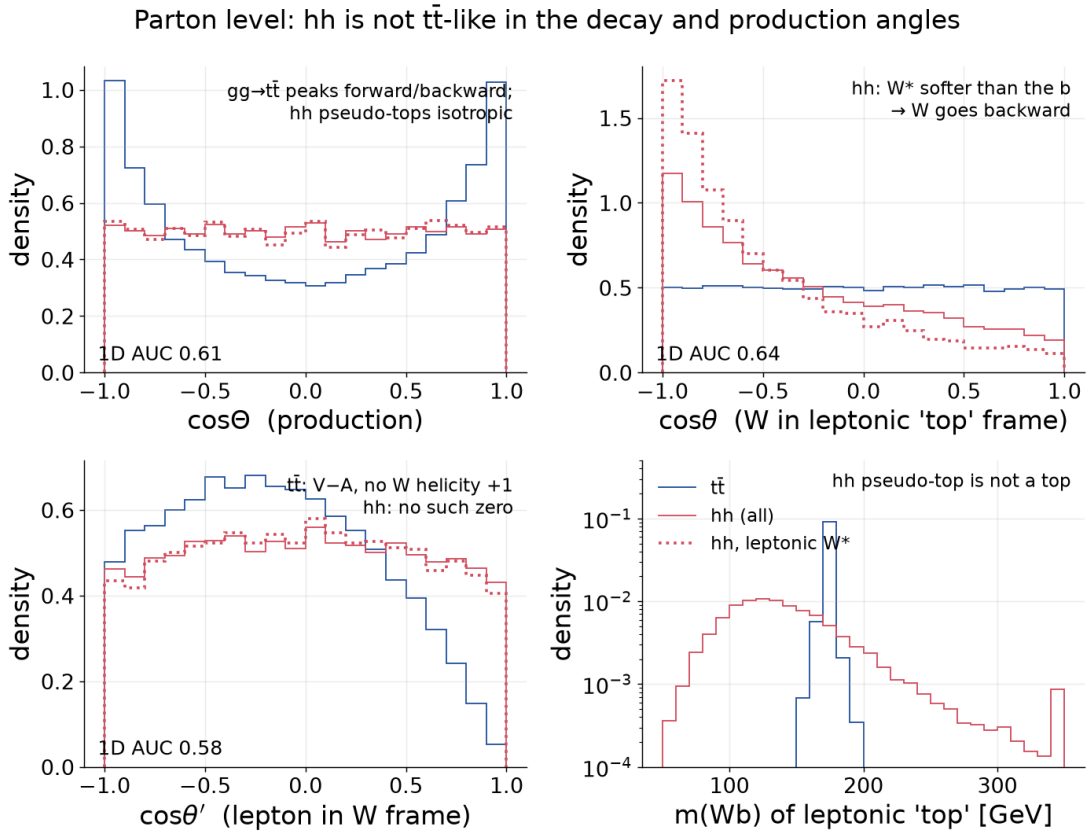


Figure 22: Parton level: what distinguishes hh from $t\bar{t}$ in the angles and the pseudo-top masses.

10.2 Step-by-step decomposition

Two independent studies agree. The first follows the physics mechanism (Figures 22–26); the second measures the loss step by step from the parton level to the full gen-level reconstruction (Table 19, Figure 23). Each step re-fits $f_{t\bar{t}}$ on the $t\bar{t}$ events of that step, scored leave-one-out; the parton-level f applied unchanged, and a supervised BDT on the ten angles, are shown alongside. An important technical point: the LHE partons are not the right reference at gen level, because Pythia's initial-state-radiation recoil kicks the whole hard system (the median ΔR between the LHE lepton and the final lepton is 0.16 for $t\bar{t}$ and 0.36 for hh). The "true" partons are therefore taken from the Pythia record, the decay products of the last-copy t, W or h after the initial-state radiation and before the final-state radiation.

The selection costs nothing, and the ISR recoil costs $f_{t\bar{t}}$ almost nothing (it costs the supervised BDT about 0.08, because the BDT's extra power at parton level relies on the exact $p_T = 0$ kinematics of the $t\bar{t}$ system). Final-state radiation costs 0.03. The largest loss, 0.06, is the choice of jets by the χ^2: on the matched subset the jets and the χ^2 cost nothing at all, so the whole loss

Table 19: AUC along the chain from parton to gen level, all selected events ($hh = 1$). LOO: $f_{t\bar{t}}$ refit at that step and scored leave-one-out; fixed: the parton-level f applied unchanged; BDT: supervised on the ten angles. Errors about ± 0.005.

step	$t\bar{t}$ / hh	$f_{t\bar{t}}$ LOO	f fixed	BDT
parton, all events	100k / 20.0k	0.650	0.643	0.881
+ gen selection	41k / 4.4k	0.659	0.690	0.866
+ ISR recoil (Pythia partons)	41k / 4.4k	0.652	0.691	0.785
+ FSR	41k / 4.4k	0.619	0.662	0.739
χ^2 jets + true ν	41k / 4.4k	0.556	0.600	0.659
full gen χ^2 (ν from MET)	41k / 4.4k	0.528	0.553	0.659

Table 20: The same on the subset of events whose four partons match distinct jets (39% of $t\bar{t}$, 12% of hh; errors about ± 0.013).

step	$f_{t\bar{t}}$ LOO	f fixed	BDT
post-ISR parton	0.599	0.664	0.743
matched jets + gen ℓ + true ν	0.608	0.659	0.753
ν p_z from m_W (true ν p_T)	0.552	0.575	0.753
ν from MET + m_W	0.564	0.579	0.741
χ^2 jets + true ν	0.605	0.655	0.706
full χ^2	0.554	0.562	0.698

is in the unmatched events, where the χ^2 has to build the system from wrong jets; for hh this is the soft W^* quarks (median p_T 27 GeV), which often make no jet. The neutrino costs 0.03 overall and 0.05 on matched events, almost all of it from the choice of the p_z solution with the m_t rule rather than from the missing-momentum resolution. Per angle (Figure 24), $\cos\Theta$ and $\cos\tilde{\theta}$ lose their separation at the jet choice, $\cos\theta'$ at the neutrino, and ISR recoil mainly costs ϕ, $\tilde{\phi}$ and Y. At the end no single angle has an AUC above 0.57.

10.3 Conclusion

The angles are the right variables for $t\bar{t}$, but at gen and reco level they are computed under the $t\bar{t}$ hypothesis, which removes exactly what makes hh different. On top of that, $f_{t\bar{t}}$ is unsupervised: it flags only events where $t\bar{t}$ is rare, and the fitted reconstruction moves hh into the region where $t\bar{t}$ is dense. The information is not gone; it sits in the dimensions that the ten angles do not span. That is the subject of the next section.

11 Adding momentum to y: the resonance masses

11.1 Motivation

The final state has six particles, i.e. 18 momentum components. They split exactly into the 10 paper variables (Y, Θ and the 8 decay angles), the 4 resonance masses $m_{W,\ell}$, $m_{W,h}$, $m_{t,\ell}$, $m_{t,h}$, and 4 production variables ($m_{t\bar{t}}$, the two components of the $t\bar{t}$ transverse momentum and an overall azimuth). For $t\bar{t}$ the four masses carry no information: each is a Breit–Wigner fixed by M and Γ, and in the narrow-width limit the matrix element factorizes into the angular part times $\prod_k \mathrm{BW}(m_k)$,

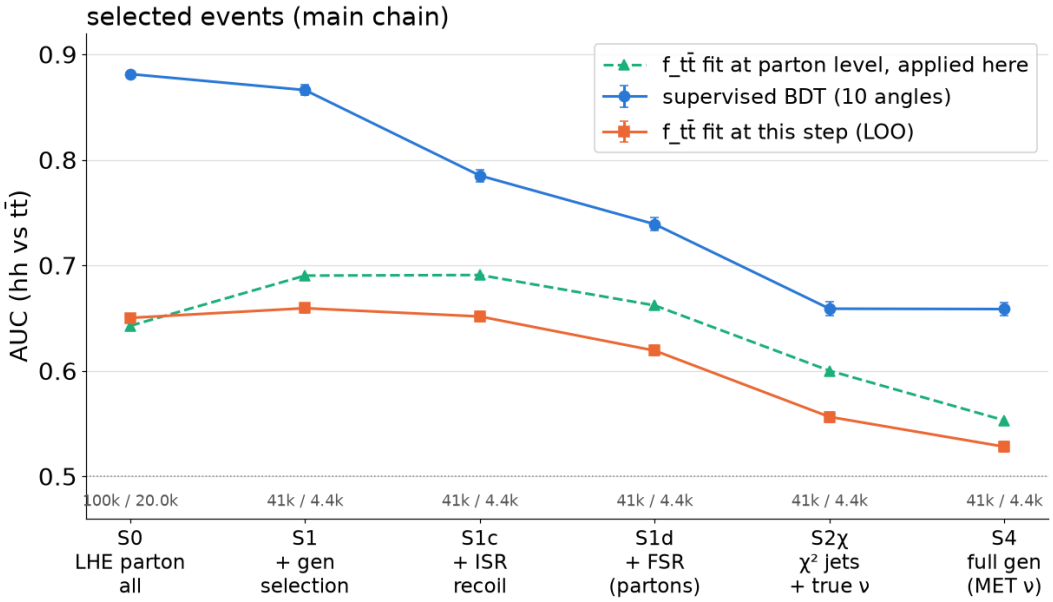


Figure 23: AUC along the chain from parton to gen level, all selected events.

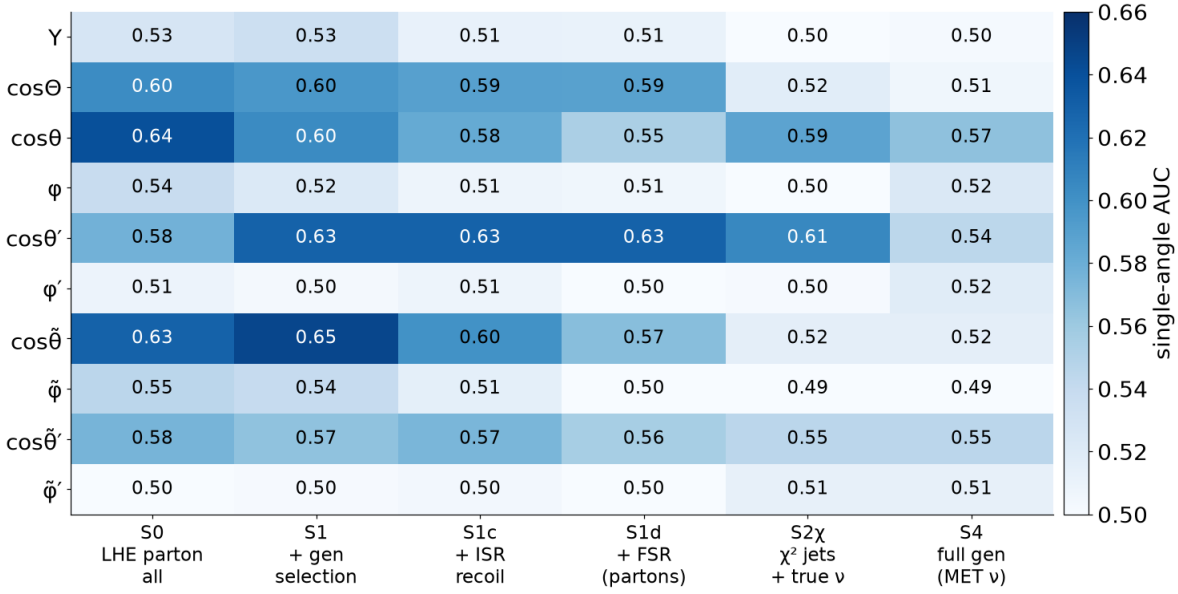


Figure 24: Single-angle AUC along the chain.

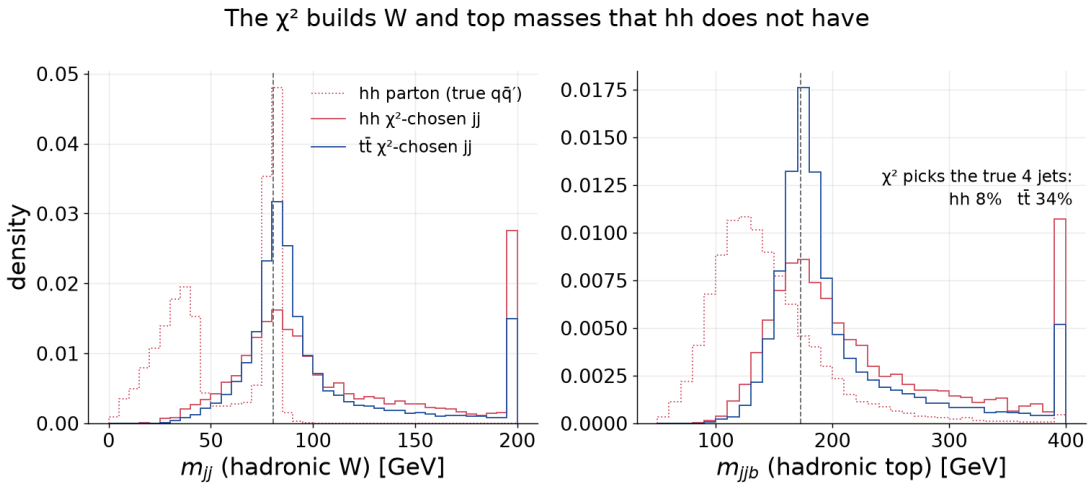


Figure 25: The χ^2 builds W and top masses that hh does not have.

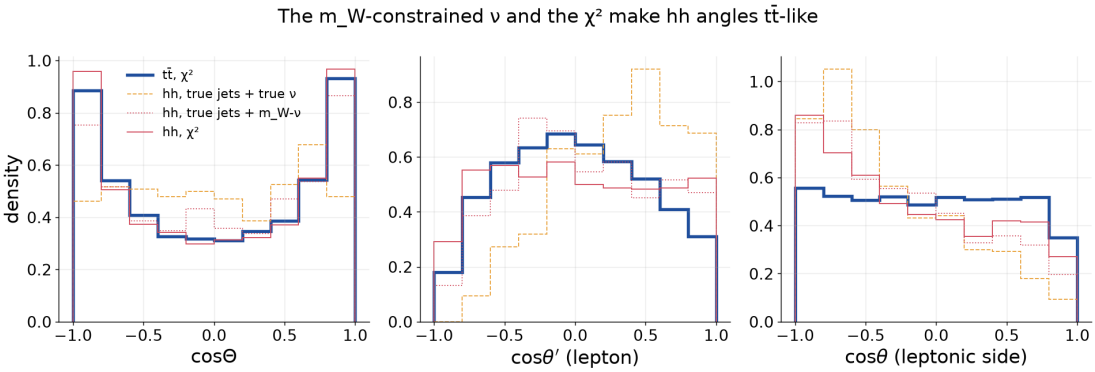


Figure 26: hh angles become $t\bar{t}$-like: $\cos\Theta$, $\cos\theta'$ and $\cos\theta$ for hh with true jets and the true neutrino, with the m_W-constrained neutrino, and after the full χ^2, against $t\bar{t}$.

so the paper could rightly drop them. hh is not on that shell: one W is off shell and its pseudo-tops sit near 135 GeV (Figure 27). The difference between hh and $t\bar{t}$ lives in exactly the dimensions that were dropped, which is the user's observation that the off-shell W makes hh a different final state from on-shell $t\bar{t}$.

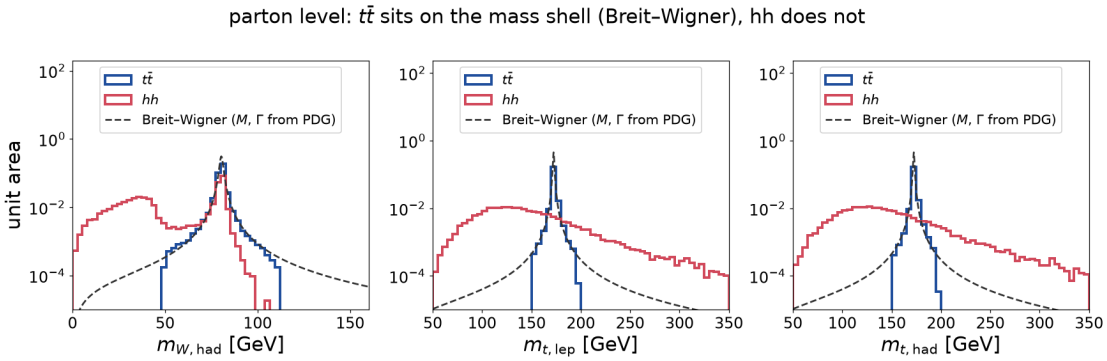


Figure 27: The dropped dimensions at parton level: $t\bar{t}$ (blue) sits on the Breit–Wigner (dashed; MadGraph cuts it at $\pm 15\Gamma$), hh (red) is spread over 100 GeV.

11.2 The extension

The extension keeps $y(\Omega)$ as it is and multiplies it by one factor per resonance mass,

$$y_{\mathrm{ext}}(\Omega, m) = y(\Omega) \prod_{k=1}^{4} \mathrm{BW}_k(m_k), \tag{3}$$

the $t\bar{t}$ density on the full phase space. In the surprisal, from which t is built, the factors simply add: $s_{\mathrm{ext}} = s(\Omega) + \sum_k \ln\left[(m_k^2 - M_k^2)^2 + M_k^2\Gamma_k^2\right] + \mathrm{const}$. At parton level M and Γ are the PDG values and nothing is fitted. At gen and reco level the detector resolution replaces Γ; the same peak shape is kept, with M and the width W taken as the median and half the interquartile range of $t\bar{t}$, i.e. two numbers per mass (Figure 28). A fitted correction to this shape, an orthonormal Legendre expansion in the Cauchy-mapped variable $u = \frac{1}{2} + \frac{1}{\pi}\arctan\frac{m-M}{W}$ with the Jacobian du/dm, was also tried (`study_salvage/salvage.py`): it changes the gen-level AUC by only +0.004, so it is not needed. At gen and reco level only three masses are available, because the neutrino solution fixes $m_{W,\ell} = m_W$ by construction.

11.3 Results

Table 21 adds the masses one at a time. At parton level the masses alone separate hh completely, since hh is never on the $t\bar{t}$ mass shell. At gen level the unsupervised score improves from 0.53 to 0.69, which is more than the angles alone achieve at parton level (0.65): the parton-level performance is regained. Table 22 is the clean version of the same result at gen and reco level, with the $t\bar{t}$ test events from the independent sample and M, W from 30k fit-sample events: reco level goes from 0.51 to 0.60. The gain is smaller at reco than at gen level because each mass separates less there (single-mass AUCs 0.574, 0.569, 0.575 against 0.628, 0.656, 0.637 at gen level), consistent with the χ^2 having more jet combinations to fit at reco level and so building $t\bar{t}$-like masses more often for hh; the reco test sample is also small (2440 $t\bar{t}$ events, errors about ± 0.01).

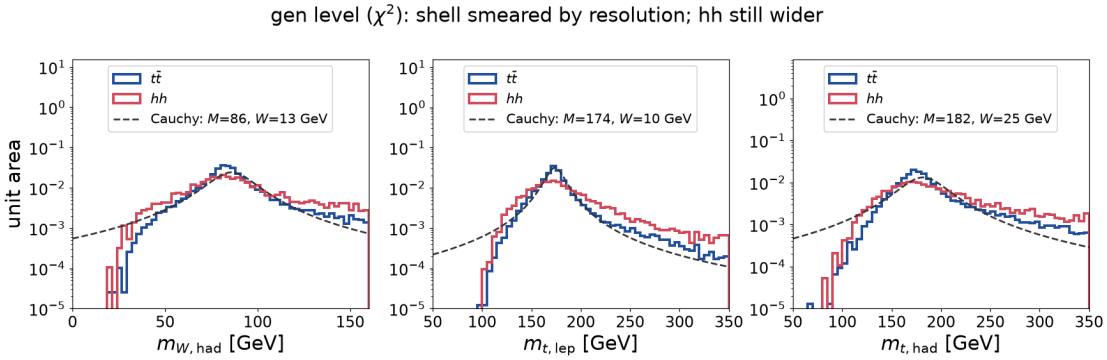


Figure 28: The same dimensions at gen level after the χ^2 reconstruction: the $t\bar{t}$ shell is widened by the resolution (dashed: the two-number model); hh is still wider, mainly in the tails.

Table 21: AUC (hh vs $t\bar{t}$), unsupervised, adding one Breit–Wigner factor at a time (test: half of 40k $t\bar{t}$ fit-sample events not used to set M, W, angles scored leave-one-out; 20k / 4.4k hh).

	parton	gen (χ^2)
angles only, $y(\Omega)$ (paper)	0.648	0.528
$+\,m_{W,\ell}$	0.786	(fixed by the ν solution)
$+\,m_{W,h}$	0.993	0.631
$+\,m_{t,\ell}$	0.999	0.683
$+\,m_{t,h}$ (all masses)	1.000	0.689

Table 22: The mass extension at gen and reco level on the independent $t\bar{t}$ sample. Widths W (gen / reco): $m_{W,h}$ 13 / 12, $m_{t,\ell}$ 10 / 16, $m_{t,h}$ 25 / 25 GeV.

level	$t\bar{t}$ / hh	angles (all)	angles ($>10\sigma$)	masses (BW)	angles ($>10\sigma$) + BW
gen	4134 / 4416	0.525	0.536	0.690	0.692
reco	2440 / 1878	0.507	0.518	0.600	0.602

11.4 What the angles add on top of the masses

In the product model the angles add nothing measurable for hh once the mass factors are in (+0.002 at gen level, +0.002 at reco level). A supervised BDT shows how much is there in principle: trained on the angles and the masses against the masses alone, it gains $+0.029 \pm 0.004$ at gen level and $+0.037 \pm 0.015$ at reco level (paired bootstrap; `study_salvage/angles_vs_bw.py`). So a little angular information survives that the product form, which assumes angles and masses independent for $t\bar{t}$, does not use; a joint fit of angles and masses could recover part of it.

11.5 How many angular terms are needed once the masses are in?

With the mass factors in, the hh separation is the same whatever the number of angular terms kept, even with none at all (Table 23): for hh the masses do all the work. Even for the angles alone, about 20 terms beat all 14 400 (Section 5.3). The recommendation is therefore $y = [\text{few-term angular } f] \times \prod_k \mathrm{BW}(m_k)$. The small angular part is worth keeping: it costs almost nothing, and it is what would catch an on-shell signal with a different spin structure, which the masses cannot see (Section 13).

Table 23: AUC with the angular density truncated by coefficient significance, with and without the mass factors.

angular terms kept	parton: terms	angles	+BW	gen: terms	angles	+BW
none (flat in the angles)	0	–	1.000	0	–	0.689
$\lvert c\rvert/\sigma_c > 40$	7	0.653	1.000	5	0.546	0.692
$\lvert c\rvert/\sigma_c > 20$	16	0.662	1.000	14	0.555	0.692
$\lvert c\rvert/\sigma_c > 10$	23	0.665	1.000	54	0.534	0.689
$\lvert c\rvert/\sigma_c > 5$	54	0.663	1.000	153	0.528	0.688
$\lvert c\rvert/\sigma_c > 3$	186	0.661	1.000	330	0.528	0.688
CV (paper method)	326	0.661	1.000	459	0.529	0.688
all	14400	0.648	1.000	14400	0.528	0.689

11.6 Realistic quark tagging

The b-tagging at reco level is the Delphes ATLAS parametrization and the χ^2 assigns the b-jets, so that part is already realistic. What the paper assumes beyond it is knowing which W jet is the down-type quark, and so far the jets were matched to the parton-level quarks, i.e. perfect up/down tagging. The only real handle is c-tagging: in $W \to cs$, 50% of the events at parton level, a c-tagged jet is up-type, so the other jet is the down-type s. Instead of a hard choice, a weight $w = P(\text{softer jet is down-type} \,|\, c\text{-tags})$ is computed by Bayes' rule from the $W \to cs$ and $W \to ud$ hypotheses, the tag efficiencies, and the prior that the softer jet is the down-type one (58% at parton level, 53% at reco level); then $y = w\, f(\Omega_{\text{down=softer}}) + (1-w)\, f(\Omega_{\text{down=harder}})$, the $t\bar{t}$ density of the event as observed, with f unchanged. Tags are emulated from the true jet flavour (the Delphes jet flavour at reco level, the quark identity at parton level) at illustrative ATLAS-like working points, ϵ_c / b mistag / light mistag of 30 / 11 / 1.4% and 45 / 30 / 8%, plus an ideal tagger.

Table 24 shows that realistic tagging costs at most 0.02 on the angles alone at parton level, nothing at reco level, and nothing at all once the masses are in, because the masses do not depend on which jet is down-type. The hadronic primed angles carry little of the hh separation anyway.

Table 24: Effect of the down-type assignment (AUC hh vs $t\bar{t}$, angular f with the terms above 10σ).

down-type assignment	label right ($t\bar{t}$)	parton: angles	+BW	reco: angles	+BW
perfect up/down tag (paper)	100%	0.670	1.000	0.518	0.602
ideal c-tag	79%	0.661	1.000	0.517	0.602
c-tag loose (45 / 30 / 8%)	65%	0.653	1.000	0.517	0.602
c-tag tight (30 / 11 / 1.4%)	64%	0.652	1.000	0.517	0.602
no tag: average both labels	–	0.650	1.000	0.517	0.602
no tag: softer jet = down	58%	0.643	1.000	0.515	0.601

12 Study A′: hh with on-shell W bosons

The off-shell W^* of $hh \to b\bar{b}\,WW^*$ is what makes the mass handle of Section 11 so effective. To see what the angles do when that handle is removed, the same $hh \to b\bar{b}\,\ell\nu\,q\bar{q}'$ chain is generated with $m_H = 170$ GeV, just above $2m_W = 160.8$ GeV, so that both W bosons are on shell ($\Gamma_H = 0.38$ GeV, shapes only), and the analysis is repeated with nothing retrained. The W Breit–Wigner factors should then go blind, the top factors should still see m_{Wb} (the b comes from the other Higgs boson), and after the χ^2 reconstruction a genuinely on-shell hadronic W should make the signal look more $t\bar{t}$-like; what is left is the angles, since a scalar decaying to WW does not give the $V-A$ shape of the top decay and the pseudo-top direction still differs.

The harder leptons raise the selection efficiency: trigger / gen / reco efficiencies of 51 / 22 / 9.4% for $hh(125)$ become 73 / 44 / 20.0% for $hh(170)$. Table 25 compares the scores. At parton level the W factors alone drop from 0.99 to 0.50 while the top factors stay at 0.99: the mass handle was mostly the pseudo-top masses, not the W^*. The angles alone go from 0.67 to 0.66 at parton level and from 0.52 to 0.55 at reco level; angles + masses from 0.69 to 0.69 at gen level and from 0.60 to 0.61 at reco level, i.e. unchanged. After reconstruction the off-shell W^* was never the handle, since the χ^2 had already pulled m_{jj} to m_W; the pseudo-top masses and, to a lesser extent, the angles are. The method therefore does not rely on the off-shell W: an hh-like signal with on-shell W bosons is separated just as well, and selected twice as often.

Table 25: $hh(125)$ and $hh(170)$ versus $t\bar{t}$: AUC / signal efficiency (%) at 1% $t\bar{t}$ efficiency, with the existing fits.

score	parton 125	parton 170	gen 125	gen 170	reco 125	reco 170
angles, all terms	0.653 / 4	0.638 / 4	0.525 / 2	0.547 / 2	0.507 / 1	0.534 / 1
angles, $> 10\sigma$	0.670 / 8	0.662 / 7	0.536 / 2	0.565 / 2	0.518 / 2	0.547 / 1
BW, W masses only	0.992 / 67	0.499 / 2	0.628 / 2	0.584 / 2	0.574 / 1	0.542 / 1
BW, top masses only	0.999 / 98	0.995 / 93	0.684 / 3	0.691 / 3	0.592 / 2	0.610 / 2
BW, all masses	1.000 / 100	0.985 / 78	0.690 / 3	0.680 / 2	0.600 / 2	0.600 / 1
angles + BW	1.000 / 100	0.986 / 80	0.692 / 2	0.685 / 2	0.602 / 2	0.606 / 1

Part II

Study B: $t\bar{t}$ versus a heavy scalar $H \to t\bar{t}$

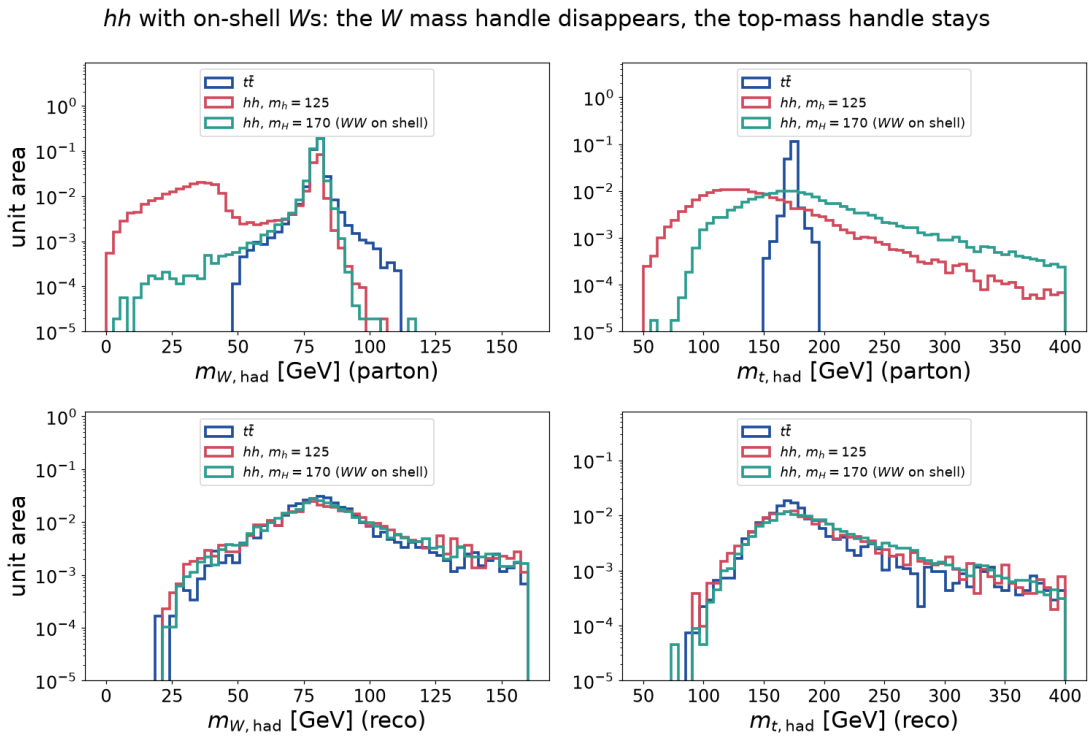


Figure 29: $m_{W,\mathrm{had}}$ and $m_{t,\mathrm{had}}$ at parton (top) and reco level (bottom) for $t\bar{t}$, $hh(125)$ and $hh(170)$.

13 A heavy scalar decaying to on-shell tops

Study A used hh, an off-shell signal: the information that separates it from $t\bar{t}$ sits in the resonance masses. Study B asks the opposite question, with new physics in the production and on-shell tops: $gg \to H \to t\bar{t} \to b\bar{b}\,\ell\nu\,q\bar{q}'$ for a heavy CP-even scalar, generated with the `heft` ggH coupling and SM-like decays (signal only, without the interference with QCD $gg \to t\bar{t}$), at $m_H = 500$ GeV ($\Gamma_H = 25$ GeV, $\Gamma/m = 5\%$ as for a heavy two-Higgs-doublet scalar) and at $m_H = 350$ GeV ($\Gamma_H = 10$ GeV), just above the $2m_t = 346$ GeV threshold. MadGraph's automatic width for this configuration is unusable (4×10^5 GeV), so the widths are set by hand; the cross sections are LO HEFT normalizations and only the shapes are used. Such a signal has exactly the masses of $t\bar{t}$, so the Breit–Wigner factors of Section 11 see nothing; it differs in the production angle (a scalar decays isotropically, whereas $gg \to t\bar{t}$ is forward–backward peaked by t-channel top exchange), in the spin correlations (a $J = 0$ state gives like-helicity tops at every mass, whereas QCD gg does so only near threshold), in the rapidity, and in $m_{t\bar{t}}$, which peaks at m_H on the bulk of the QCD spectrum. At threshold $\cos\Theta$ is isotropic for QCD as well (S-wave), so only the spin correlations and $m_{t\bar{t}}$ remain; this is the kinematic region of the CMS threshold excess, where a pseudoscalar ${}^1S_0$ state is indistinguishable from QCD gg in its spin correlations while a scalar ${}^3P_0$ differs in the transverse ones. Two uses of $f_{t\bar{t}}$ are tested: as an anomaly score, and its coefficients as a bump hunt in angle space.

13.1 What differs at parton level

Table 26 gives, for each of the ten angles and $m_{t\bar{t}}$, the AUC of a one-dimensional likelihood ratio (a 40-bin histogram ratio fitted on half of the events and tested on the other half, so that non-monotonic differences count). For both signals the production angle and $m_{t\bar{t}}$ carry the difference; the single decay angles show nothing, as they must, since spin correlations change the joint distribution and not the one-dimensional marginals (Figure 30). The $H(350)$ sample has a median $m_{t\bar{t}}$ of 384 GeV because of the width and the phase space, so it sits above the threshold region where the QCD production angle flattens.

Table 26: One-dimensional likelihood-ratio AUC of each variable, $H \to t\bar{t}$ versus $t\bar{t}$ at parton level.

variable	$H(500)$	$H(350)$
Y	0.515	0.499
$\cos\Theta$	0.615	0.610
$\cos\theta$	0.512	0.497
ϕ	0.501	0.498
$\cos\theta'$	0.498	0.504
ϕ'	0.506	0.499
$\cos\tilde{\theta}$	0.501	0.500
$\tilde{\phi}$	0.506	0.501
$\cos\tilde{\theta}'$	0.493	0.497
$\tilde{\phi}'$	0.503	0.502
$m_{t\bar{t}}$	0.880	0.734

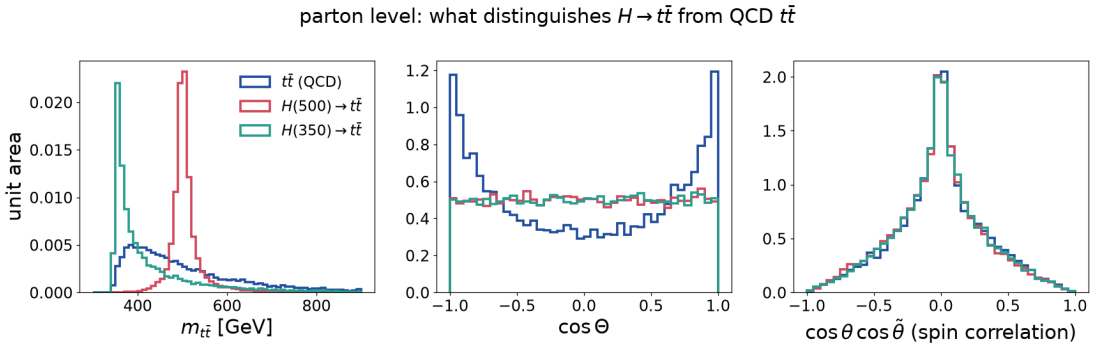


Figure 30: Parton level: $m_{t\bar{t}}$, $\cos\Theta$ and the product $\cos\theta\cos\tilde{\theta}$ (a spin-correlation proxy) for QCD $t\bar{t}$, $H(500)$ and $H(350)$.

13.2 $f_{t\bar{t}}$ as an anomaly score

The masses see nothing, as they must for an on-shell signal. The angles alone give a weak but real separation, 0.55 for $H(500)$ and 0.57 for $H(350)$ at parton level, and the mass factors *dilute* it, because the summed surprisal is dominated by the dimensions with the largest spread; for an on-shell signal the angular part alone is the right score. $m_{t\bar{t}}$ as a one-dimensional $t\bar{t}$ density helps little although it is the most different variable: the resonance sits where the $t\bar{t}$ density is high, and a score that looks for *low* density finds rarities, not excesses. A dedicated supervised search does far better because it knows the signal. A vector resonance would not change this picture for

Table 27: $H(500)$ versus $t\bar{t}$: AUC of each score and the signal efficiency (%) at $t\bar{t}$ efficiency 1% / 10% (1 / 10 would be no enrichment). Reference: a BDT trained on this signal with angles, masses and $m_{t\bar{t}}$, what a dedicated search could do.

score	parton AUC	ϵ_S (%)	gen AUC	ϵ_S (%)	reco AUC	ϵ_S (%)
angles, all terms	0.561 ± 0.004	0.8 / 11	0.559 ± 0.006	1.3 / 12	0.560 ± 0.006	0.8 / 12
angles, $> 10\sigma$	0.555 ± 0.004	0.7 / 10	0.562 ± 0.006	1.4 / 12	0.565 ± 0.007	0.8 / 12
BW masses	0.508 ± 0.003	1.0 / 10	0.541 ± 0.005	2.3 / 15	0.533 ± 0.007	2.2 / 13
$m_{t\bar{t}}$ density	0.557 ± 0.004	0.1 / 1	0.492 ± 0.006	1.1 / 9	0.483 ± 0.007	0.9 / 8
angles + BW	0.514 ± 0.003	1.1 / 10	0.552 ± 0.005	2.2 / 15	0.547 ± 0.007	2.2 / 14
angles + $m_{t\bar{t}}$	0.529 ± 0.004	0.1 / 1	0.522 ± 0.006	1.4 / 10	0.515 ± 0.007	1.1 / 9
angles + BW + $m_{t\bar{t}}$	0.493 ± 0.003	0.9 / 9	0.533 ± 0.006	2.3 / 15	0.527 ± 0.007	2.4 / 13
dedicated supervised BDT (reference)	0.894	–	0.693	–	0.669	–

Table 28: $H(350)$ versus $t\bar{t}$: AUC of each score and the signal efficiency (%) at $t\bar{t}$ efficiency 1% / 10% (1 / 10 would be no enrichment). Reference: a BDT trained on this signal with angles, masses and $m_{t\bar{t}}$, what a dedicated search could do.

score	parton AUC	ϵ_S (%)	gen AUC	ϵ_S (%)	reco AUC	ϵ_S (%)
angles, all terms	0.579 ± 0.004	1.0 / 13	0.549 ± 0.005	1.6 / 12	0.542 ± 0.008	0.7 / 11
angles, $> 10\sigma$	0.572 ± 0.004	1.3 / 12	0.550 ± 0.005	1.2 / 11	0.553 ± 0.008	0.9 / 11
BW masses	0.512 ± 0.004	1.1 / 11	0.537 ± 0.005	1.2 / 13	0.509 ± 0.007	1.2 / 11
$m_{t\bar{t}}$ density	0.467 ± 0.003	0.4 / 4	0.505 ± 0.005	0.8 / 11	0.495 ± 0.006	0.9 / 7
angles + BW	0.520 ± 0.004	1.2 / 11	0.546 ± 0.005	1.1 / 13	0.522 ± 0.007	1.1 / 11
angles + $m_{t\bar{t}}$	0.511 ± 0.004	0.6 / 5	0.526 ± 0.006	1.2 / 11	0.522 ± 0.007	1.0 / 11
angles + BW + $m_{t\bar{t}}$	0.511 ± 0.004	1.3 / 10	0.538 ± 0.005	1.3 / 13	0.513 ± 0.007	1.4 / 11
dedicated supervised BDT (reference)	0.764	–	0.638	–	0.577	–

the score: from $q\bar{q}$ it would give $1 + \cos^2\Theta$, closer to the QCD shape than a flat distribution, and opposite-helicity tops instead of like-helicity ones, i.e. a different sign of the spin correlation with the same magnitude; chiral couplings would add a net top polarization, a $1 + 0.4P\cos\theta$ modulation for the score but a pull on terms that are zero in the SM for the test below.

13.3 The OSDE coefficients as a bump hunt in angle space

The same fit can be used differently. The coefficients fitted on data, $\hat{c}_{\text{data}}$, are estimates of the same quantities as the $t\bar{t}$ reference coefficients $\hat{c}_{t\bar{t}}$ from the fit sample, with known statistical errors; a signal shows as a difference on specific terms, which an excess produces just as well as a rarity, and the terms say what the physics is. Here the "data" is the independent $t\bar{t}$ sample (10k events) with a signal fraction ϵ added, and the pull of each of the 14 400 real degrees of freedom is $(\hat{c}_{\text{data}} - \hat{c}_{t\bar{t}})/\sqrt{\sigma^2_{\text{data}} + \sigma^2_{t\bar{t}}}$ (a conjugate pair of complex coefficients carries two real degrees of freedom, each with half the variance). At $\epsilon = 0$ the pulls have standard deviation 0.99 and the global χ^2 over all terms gives $z = -1.1$: the two $t\bar{t}$ samples agree and the errors are right. The χ^2 is then evaluated over all terms and over the terms of each physics category (production angle, rapidity, W helicity, top polarization, spin correlation and the polarization–helicity interference), with $z = (\chi^2 - \text{ndf})/\sqrt{2\,\text{ndf}}$, and the largest single pull is kept as a local test (with 14 400 trials, 5σ global corresponds to $|\text{pull}| \approx 5.3$; the largest of 14 400 null pulls is about 3.9). Figure 31 shows the result against the signal fraction. With 10k events no category χ^2 responds up to $\epsilon = 10\%$: a single term carries the signal and the rest of its block is noise. That term is $P_2(\cos\Theta)$, the isotropy of the scalar decay, whose pull at $\epsilon = 10\%$ is 3.6 for $H(500)$ and 4.0 for $H(350)$, just at the noise

ceiling. The pull of a term grows as $\epsilon\sqrt{N}$, so 5σ needs $\epsilon\sqrt{N} \approx 16$ for $H(500)$: $\epsilon \approx 1.6\%$ with 10^6 events and 0.16% with 10^8 (the HL-LHC ℓ+jets $t\bar{t}$ sample). For comparison, a bump hunt in $m_{t\bar{t}}$ ($S/\sqrt{B}$ in a $\pm\Gamma_H$ window holding about 15% of $t\bar{t}$) is roughly seven times more sensitive to this signal. The angular coefficients are therefore a characterization tool for an on-shell resonance (its spin, from which terms move), not the discovery tool; the discovery tool is $m_{t\bar{t}}$.

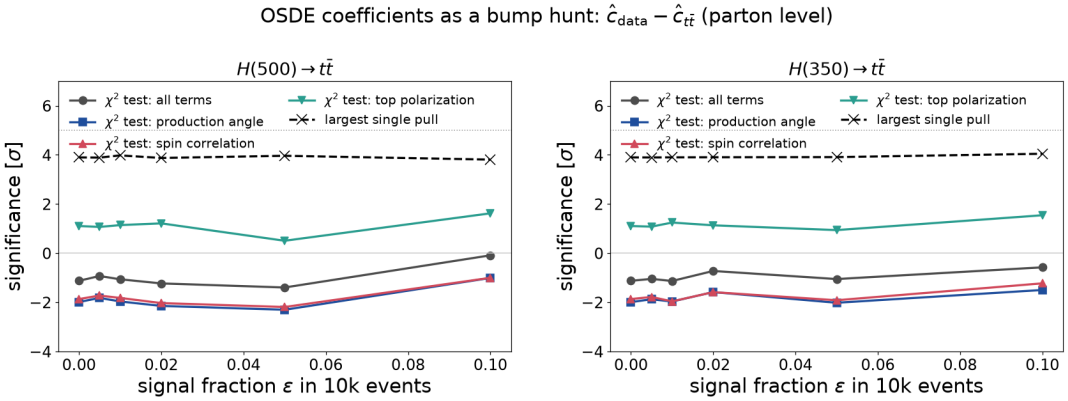


Figure 31: Significance of the coefficient test against the signal fraction in 10k events, for the χ^2 over all terms and over physics categories, and for the largest single pull.

Table 29: $H(500)$ at $\epsilon = 10\%$: χ^2 per physics category.

terms	χ^2 / ndf	z
all terms	14385 / 14400	-0.1
production angle	11367 / 11520	-1.0
rapidity	11488 / 11520	-0.2
W helicity	13974 / 14000	-0.2
top polarization	3508 / 3375	+1.6
spin correlation	10876 / 11025	-1.0
pol. x W helicity interference	13493 / 13500	-0.0

Table 30: $H(500)$ at $\epsilon = 10\%$: the terms that respond. The last column is the term's significance in the $t\bar{t}$ reference; a large pull on a term that is zero in $t\bar{t}$ is new physics in a channel the SM does not have.

term	physics	|pull|	$\lvert\hat{c}\rvert/\sigma$ in $t\bar{t}$
$M^{11}_{0,-1} \times \tilde{M}^{12}_{1,1} \times P_2(\cos\Theta)$	$t\bar{t}$ spin correlation; top pol. x W helicity interference (lep); top pol. x W helicity interference (had); production angle	3.8	2
$M^{12}_{-1,1} \times \tilde{M}^{01}_{0,0} \times P_1(\cos\Theta) \times h_2(Y)$	top pol. x W helicity interference (lep); W helicity (had); production angle; rapidity	3.8	2
$M^{10}_{-1,0} \times \tilde{M}^{12}_{1,-1} \times P_1(\cos\Theta) \times h_4(Y)$	$t\bar{t}$ spin correlation; top polarization (lep, transverse); top pol. x W helicity interference (had); production angle; rapidity	3.7	1
$M^{01}_{0,0} \times \tilde{M}^{12}_{-1,-1} \times P_2(\cos\Theta) \times h_4(Y)$	W helicity (lep); top pol. x W helicity interference (had); production angle; rapidity	3.7	2
$\tilde{M}^{12}_{1,-1} \times P_2(\cos\Theta)$	top pol. x W helicity interference (had); production angle	3.6	0
$M^{12}_{-1,-1} \times \tilde{M}^{11}_{-1,0} \times P_2(\cos\Theta) \times h_4(Y)$	$t\bar{t}$ spin correlation; top pol. x W helicity interference (lep); top pol. x W helicity interference (had); production angle; rapidity	3.6	1
$P_2(\cos\Theta)$	production angle	3.6	115
$M^{11}_{-1,1} \times \tilde{M}^{02}_{0,0}$	top pol. x W helicity interference (lep); W helicity (had)	3.6	2

13.4 New physics in the top decay: a right-handed W

The complementary case is new physics in the decay vertex, which changes the angles themselves: a right-handed W helicity fraction $F_R > 0$ fills the region $\cos\theta' \to +1$ where the SM $(V-A)$ decay density vanishes, the textbook situation for a density method. It is constrained to $F_R \lesssim 0.03$ by the W helicity and single-top measurements, so it is a benchmark rather than a discovery channel.

Table 31: $H(350)$ at $\epsilon = 10\%$: χ^2 per physics category.

terms	χ^2 / ndf	z
all terms	14302 / 14400	-0.6
production angle	11292 / 11520	-1.5
rapidity	11450 / 11520	-0.5
W helicity	13871 / 14000	-0.8
top polarization	3502 / 3375	+1.5
spin correlation	10842 / 11025	-1.2
pol. x W helicity interference	13379 / 13500	-0.7

Table 32: $H(350)$ at $\epsilon = 10\%$: the terms that respond. The last column is the term's significance in the $t\bar{t}$ reference; a large pull on a term that is zero in $t\bar{t}$ is new physics in a channel the SM does not have.

term	physics	|pull|	$\lvert\hat{c}\rvert/\sigma$ in $t\bar{t}$
$P_2(\cos\Theta)$	production angle	4.0	115
$M^{11}_{-1,1} \times \bar{M}^{11}_{1,-1} \times h_4(Y)$	$t\bar{t}$ spin correlation; top pol. x W helicity interference (lep); top pol. x W helicity interference (had); rapidity	3.8	1
$M^{10}_{-1,0} \times \bar{M}^{12}_{1,-1} \times P_1(\cos\Theta) \times h_4(Y)$	$t\bar{t}$ spin correlation; top polarization (lep, transverse); top pol. x W helicity interference (had); production angle; rapidity	3.8	1
$M^{12}_{-1,1} \times \bar{M}^{01}_{0,0} \times P_1(\cos\Theta) \times h_2(Y)$	top pol. x W helicity interference (lep); W helicity (had); production angle; rapidity	3.7	2
$M^{01}_{0,0} \times \bar{M}^{11}_{-1,1} \times P_2(\cos\Theta) \times h_3(Y)$	W helicity (lep); top pol. x W helicity interference (had); production angle; rapidity	3.7	2
$M^{10}_{-1,0} \times \bar{M}^{12}_{-1,0} \times P_3(\cos\Theta)$	$t\bar{t}$ spin correlation; top polarization (lep, transverse); top pol. x W helicity interference (had); production angle	3.7	1
$M^{01}_{0,0} \times \bar{M}^{12}_{-1,-1} \times P_2(\cos\Theta) \times h_4(Y)$	W helicity (lep); top pol. x W helicity interference (had); production angle; rapidity	3.6	2
$\bar{M}^{12}_{-1,-1} \times P_2(\cos\Theta)$	top pol. x W helicity interference (had); production angle	3.6	0

The signal is emulated by reweighting the independent $t\bar{t}$ sample with the parton-level W-decay distributions on both sides, $\rho(c) = \frac{3}{4}(1-c^2)F_0 + \frac{3}{8}(1-c)^2F_L + \frac{3}{8}(1+c)^2F_R$ with F_0 kept and $F_L \to F_L - F_R$ ($F^{\rm SM} = (0.703, 0.296, 0.001)$), so that the gen and reco scores of the same events carry the parton weights; the coefficients are fitted with the event weights, $\hat{c} = \sum wB^* / \sum w$, with the weighted-variance error.

Table 33 gives the anomaly score. The tail is real but weak: at parton level $F_R = 0.3$ gives AUC 0.60 and puts 5% of the signal into the 1% $t\bar{t}$ tail, a factor 5 enrichment; at $F_R = 0.05$ only 1.6%; at reco level the smeared $\cos\theta'$ leaves nothing, and the mass factors dilute the score as they did for H. The coefficient test (Table 34, Figure 32) is a different story: the $\cos\theta'$ term of the leptonic W, M^{01}_{00}, whose coefficient is $\langle\cos\theta'\rangle = (F_R - F_L)/2$, has a pull of 8.6 at $F_R = 0.05$ and 15.2 at $F_R = 0.1$ with 10k parton-level events, and the hadronic side follows; at gen level the pull is 6.2 at $F_R = 0.1$ with 4.1k events and at reco level 7.4 at $F_R = 0.3$ with 2.4k events, so the smearing costs sensitivity but the responding term is the same. The χ^2 over the whole "W helicity" category is diluted by its thousands of product terms and moves only at $F_R = 0.3$; the single term is the sensitive statistic. Since the pull grows as $F_R\sqrt{N}$, 5σ at the current limit $F_R = 0.02$ needs about 23k parton-level events, more after reconstruction. This is the W helicity measurement in disguise, with the physics labelled by the term. The contrast with $H \to t\bar{t}$ is instructive: new physics in the decay changes the shape of the most significant terms of f and gives large pulls on a few terms, whereas a resonance in the bulk changes the production-angle coefficient by a few per cent. The score finds signals in SM-forbidden regions when they are large; the coefficients find small deviations in the SM terms. Both are model-agnostic; the second is the sensitive one for new physics that is still allowed.

13.5 Conclusions of Study B

As an anomaly *score*, $f_{t\bar{t}}$ is weak for an on-shell resonance, scalar or vector, because the signal sits where $t\bar{t}$ is dense. As a test in *coefficient space* the same fit is calibrated and physical, but for this signal it is carried by one term, $P_2(\cos\Theta)$, whose significance grows as $\epsilon\sqrt{N}$ and stays well below

Table 33: Right-handed W: AUC / signal efficiency (%) at 1% $t\bar{t}$ efficiency of the anomaly score.

score		parton	gen	reco
angles ($> 10\sigma$)	$F_R = 0.02$	0.507 / 1.2	0.501 / 1.0	0.501 / 1.1
	$F_R = 0.05$	0.517 / 1.6	0.504 / 1.1	0.501 / 1.1
	$F_R = 0.1$	0.534 / 2.2	0.507 / 1.1	0.503 / 1.3
	$F_R = 0.3$	0.600 / 4.9	0.522 / 1.3	0.505 / 1.6
angles + BW	$F_R = 0.02$	0.501 / 1.0	0.500 / 1.0	0.501 / 1.0
	$F_R = 0.05$	0.502 / 1.0	0.501 / 1.0	0.503 / 1.0
	$F_R = 0.1$	0.505 / 1.0	0.502 / 1.0	0.505 / 1.0
	$F_R = 0.3$	0.513 / 1.0	0.507 / 0.8	0.513 / 0.8

Table 34: Right-handed W: coefficient test, pull of the $\cos\theta'$ term M_{00}^{01} on the leptonic / hadronic side and the largest single pull, per level (<4: not among the leading terms).

	parton ℓ	had	max	gen ℓ	had	max	reco ℓ	had	max
$F_R = 0$	<4	<4	3.9	<4	<4	4.0	<4	<4	4.6
$F_R = 0.01$	<4	<4	3.9	<4	<4	4.0	<4	<4	4.5
$F_R = 0.02$	3.9	<4	4.0	<4	<4	4.0	<4	<4	4.4
$F_R = 0.05$	8.6	6.9	8.6	<4	<4	4.0	<4	<4	4.2
$F_R = 0.1$	15.2	13.6	15.2	6.2	<4	6.2	<4	<4	4.1
$F_R = 0.3$	26.9	26.1	26.9	12.1	7.3	12.1	7.4	4.9	7.4

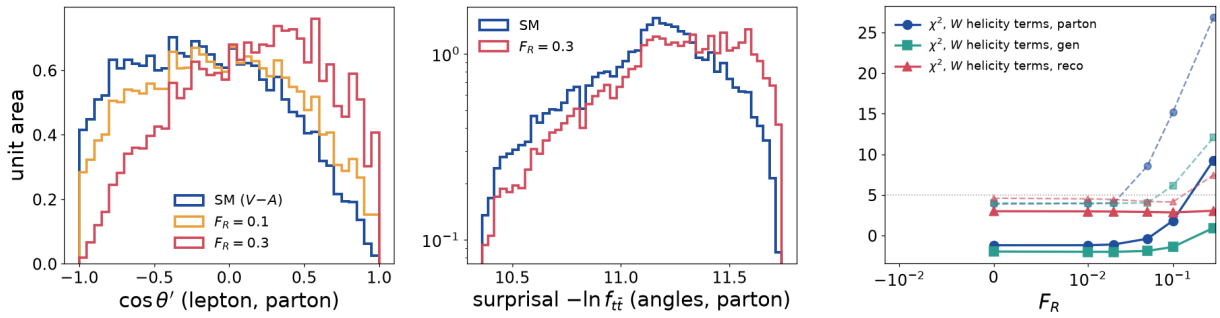


Figure 32: Right-handed W: $\cos\theta'$ at parton level for the SM and $F_R = 0.1$, 0.3 (left); the surprisal tail (middle); the significance [σ] of the W-helicity coefficient test versus F_R at the three levels (right; solid: the χ^2 of the "W helicity" category as z, dashed: the largest single pull, dotted: 5σ).

that of a bump hunt in $m_{t\bar{t}}$; the spin-correlation terms do not respond at the signal fractions tested, at 500 GeV or at threshold. The natural next steps are the test at particle and detector level with the extended basis, a pseudoscalar A (the CMS threshold case), a vector with chiral couplings, and realistic signal fractions from $\sigma \times \mathcal{B}$.

Part III
Study C: towards the detector and the trigger

14 Pileup

All results above are for zero pileup. The reco-level study is repeated at mean pileup $\langle\mu\rangle = 50$ and 200 with the Delphes ATLAS pileup card (pileup-subtracted jets, the same object and selection cuts), with the reco-level $f_{t\bar{t}}$ refitted at each pileup level and the mass extension applied. Gen level is unaffected by pileup by construction.

Table 35: Reco-level results versus pileup.

	$\langle\mu\rangle = 0$	$\langle\mu\rangle = 50$	$\langle\mu\rangle = 200$
$t\bar{t}$ fit / test events	10217 / 1965	13763 / 2755	15973 / 3221
hh events	1515	2677	3906
reco efficiency $t\bar{t}$ / hh	19.7% / 7.6%	27.6% / 13.4%	32.2% / 19.5%
mean jets ($p_T > 25$ GeV), $t\bar{t}$	3.73	4.43	7.75
all 4 jets correct, $t\bar{t}$	17.5%	5.8%	1.1%
ν p_z 68% half-width [GeV]	65	92	124
W of $m_{W,h}$ / $m_{t,\ell}$ / $m_{t,h}$ [GeV]	14 / 15 / 27	14 / 22 / 28	13 / 33 / 27
AUC angles, all terms	0.494	0.482	0.474
AUC angles, $> 10\sigma$	0.524	0.497	0.476
AUC masses (BW)	0.616	0.540	0.499
AUC angles ($> 10\sigma$) + BW	0.617	0.538	0.497

From $\langle\mu\rangle = 0$ to 200 the reco selection efficiency of $t\bar{t}$ goes from 20% to 32% (pileup jets pass the $\geq$ 4-jet cut; the card has no pileup-jet rejection), the mean jet multiplicity from 3.7 to 7.8, the fraction of events with all four χ^2 jets correct from 18% to 1.1%, and the neutrino p_z half-width from 65 to 124 GeV. The hh separation of the masses goes from AUC 0.616 to 0.499 and that of angles + masses from 0.617 to 0.497; the gen-level check is 0.698, 0.698, 0.698 at the three pileup levels, as it must be for paired events that do not see pileup at gen level.

15 A lookup-table version for the Level-1 trigger

Can the angles + Breit–Wigner score run in the ATLAS Phase-II Global Trigger, on an AMD Versal, within 5% of the device and 10 μs? The study (`study_l1/design.md`, with a bit-true fixed-point emulation `study_l1/emulate.py` run on the real reco-level samples) says yes, with large margins. The inputs are L0 objects, the six leading jets (p_T, η, ϕ, massless), one lepton and the missing transverse momentum, quantized like the Run-3 trigger objects and without b-tagging. The expensive part is not the lookup tables but the χ^2 jet assignment, 180 assignments times two neutrino solutions; it becomes cheap because for massless objects every hypothesis mass is a sum of

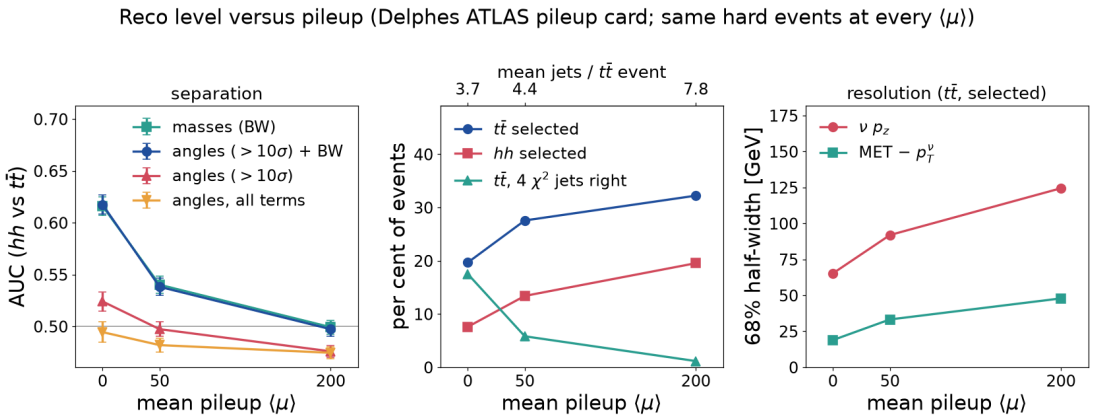


Figure 33: AUC, selection efficiencies and resolutions at reco level versus $\langle\mu\rangle$.

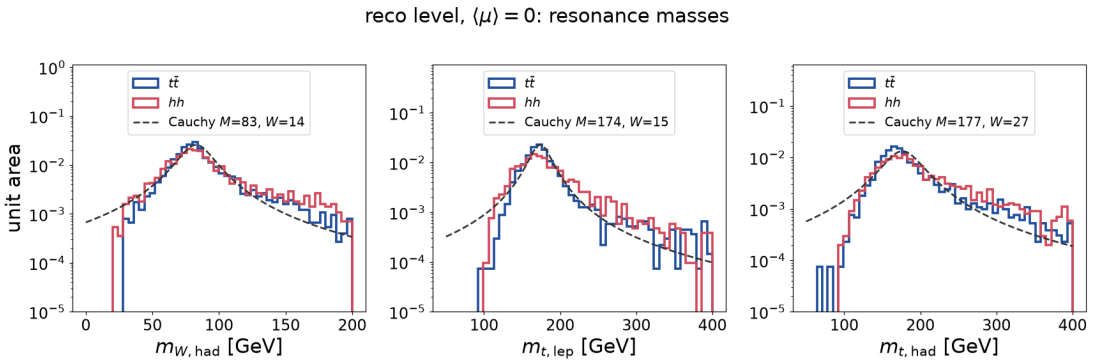


Figure 34: Resonance masses at reco level for $\langle\mu\rangle = 0$ ($t\bar{t}$ and hh, with the two-number Cauchy model).

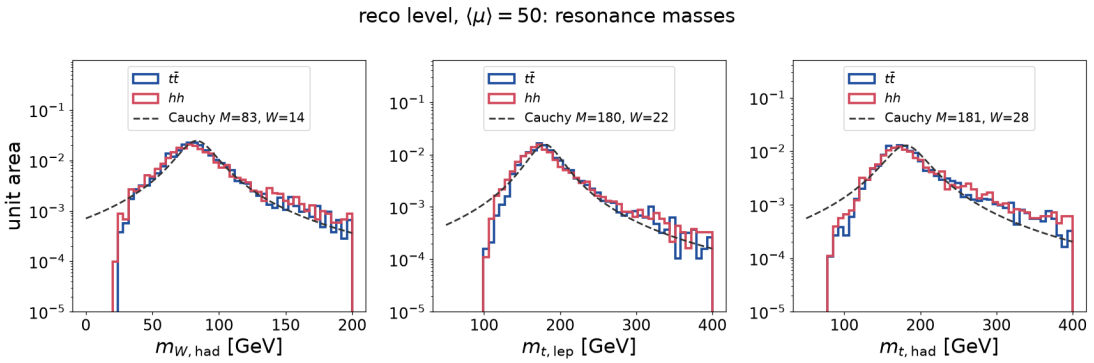


Figure 35: Resonance masses at reco level for $\langle\mu\rangle = 50$ ($t\bar{t}$ and hh, with the two-number Cauchy model).

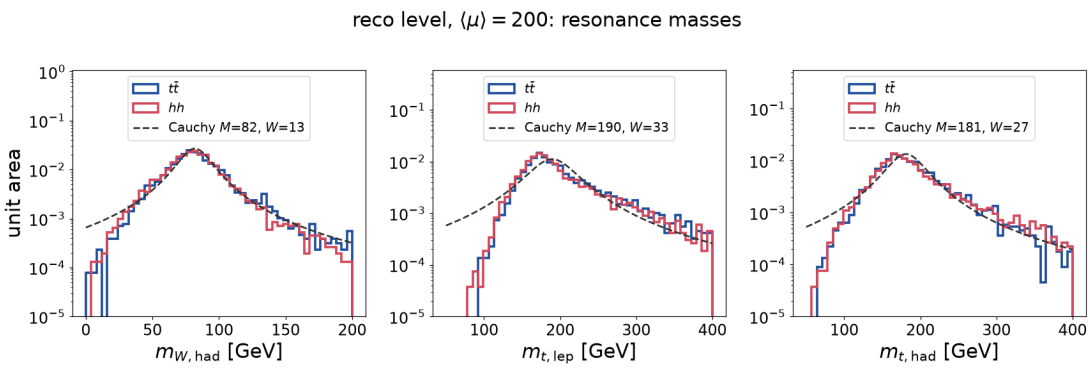


Figure 36: Resonance masses at reco level for $\langle\mu\rangle = 200$ ($t\bar{t}$ and hh, with the two-number Cauchy model).

shared pairwise invariants $m_{ij}^2 = 2p_i{\cdot}p_j$, so the whole combinatorics needs 35 four-DSP dot products, 47 small χ^2 ROMs indexed by m^2 (no square roots), 360 adders and a comparator tree. The ten angles need no Lorentz boosts: they follow from closed-form dimensionless expressions (aberration, ratios of the same invariants, perpendicular components), checked against `plotting/angles.py` to 10^{-10}. The density (31 kept terms) and the mass factors are one-dimensional ROMs. The floating-point chain reproduces the offline reference (angles + BW AUC 0.602 against 0.602).

The recommended configuration, six jets, a 16-bit datapath and 256-entry ROMs, fully parallel, costs on a Versal Premium VP1802 (the FPGA of the Global Common Module) 0.83% of the LUTs, 1.45% of the DSPs and 0.10% of the BRAM for the masses-only score, with a latency of 0.43 μs at 240 MHz, and 1.21% / 3.36% / 0.26% and 0.79 μs with the angular term; both are inside the 1.2 μs per-event budget of a trigger node. Fixed point costs nothing in AUC for widths of 14 bits or more and ROM depths of 64 or more; the hh-like tail survives quantization (at 5–10% $t\bar{t}$ acceptance, at least 95% of the selected events are the same as in floating point, and 91% of the events get the same χ^2 assignment). What limits the physics at L1 is the input: the AUC falls from the offline 0.602 to 0.583 because there is no b-tag; restricting to six jets is free, four jets would cost another 0.018. The angles add $+0.0004 \pm 0.0022$, so the masses-only design is the recommendation. The resource and latency numbers are analytic, since no Vivado is available here; the document gives the validation path with Vitis HLS and hls4ml.

16 Open questions and next steps

- **Conventions to confirm with the paper.** $y_0 = 2.5 \min y$ was inferred from the paper's figure; the sign convention in the paper's text and in its caption disagree.
- **Neutrino.** Solving the neutrino with the $m_{t,\ell}$ constraint instead of m_W would free $m_{W,\ell}$, which is the off-shell W^* in half of the hh events, and would remove the largest single loss on correctly reconstructed events.
- **No single hypothesis.** Instead of committing to one χ^2 assignment, evaluate $y = \sum_a w_a\, f_{t\bar{t}}(\Omega_a, m_a)$ summed over jet assignments and neutrino solutions, the density of the observed final state under $t\bar{t}$ in the spirit of the matrix-element method; this removes the χ^2's pull of hh onto the $t\bar{t}$ shell.
- **Joint fit.** Fit angles and masses jointly rather than as a product, by adding Legendre-in-u

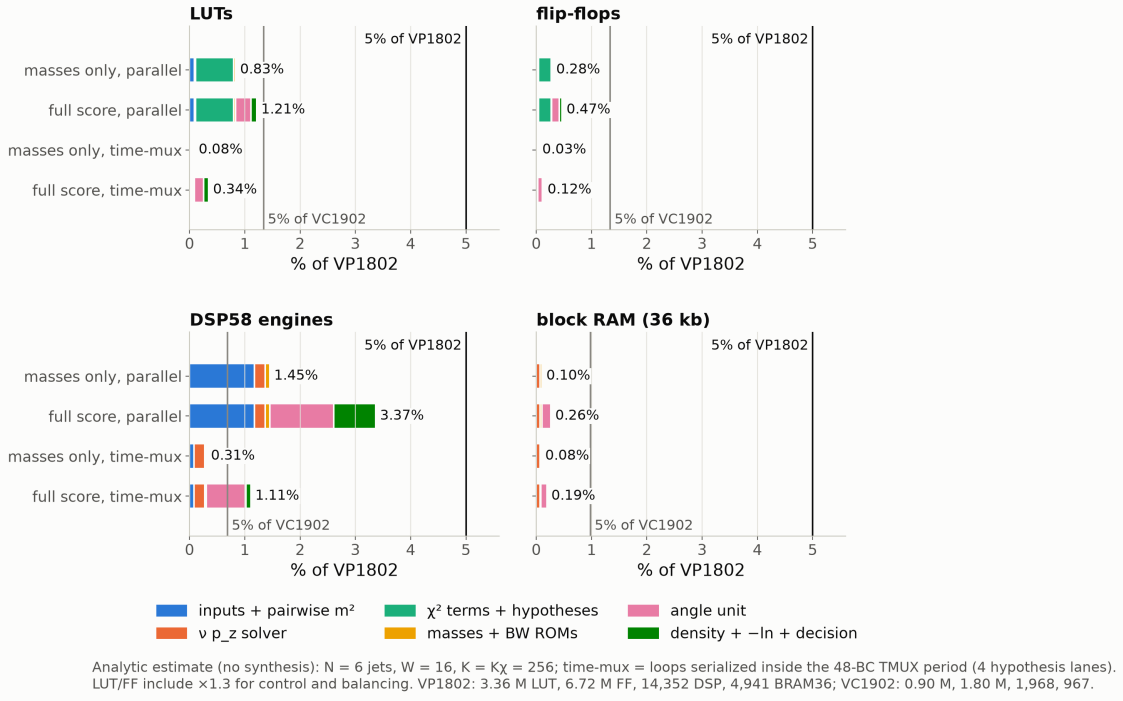


Figure 37: Resources by block as a fraction of a VP1802, against the 5% budget.

mass blocks to the extended block basis; with $m_{b\bar{b}}$ included the product form already fails because angles and masses become correlated.

- **Production variables.** $m_{t\bar{t}}$ and the $t\bar{t}$ transverse momentum are the remaining dimensions; a resonance such as $H \to t\bar{t}$ shows up in $m_{t\bar{t}}$.
- **Binning and defaults.** Equiprobable-contour binning of t; the truncated or extended f as the default; an f_{hh} for A–B testing.

A Code and reproduction

Everything is in `/Users/TMHONG/Claude-code/ttbar_bblvqq` (git repository on `main`).

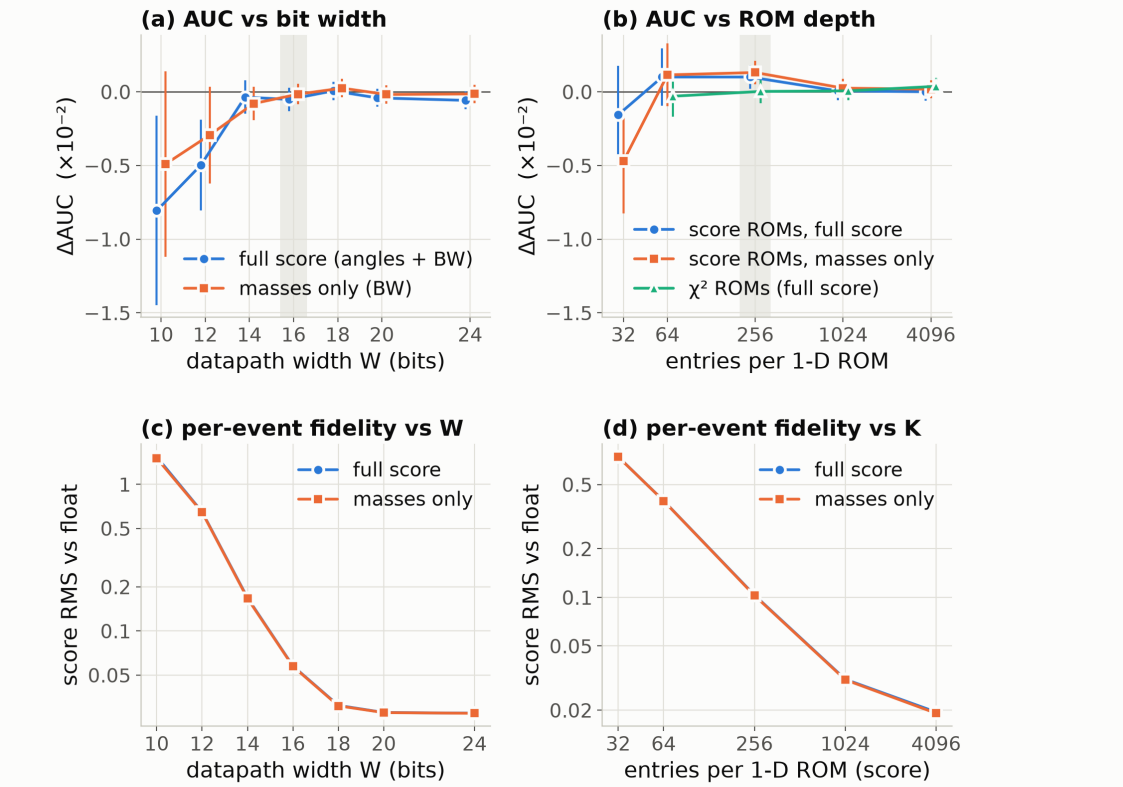
`setup_tools.sh` installs MadGraph5_aMC 3.6.7, Pythia8, Delphes 3.5.1 (built with CMake) and the Python environment.

`generate_samples.py` MadGraph → Pythia8 → Delphes production; `--process ttbar|hh|Htt`.

`plotting/ io.py` (LHE and Delphes readers), `physics.py` (objects, selection, χ^2 reconstruction), `angles.py` (the ten angles), `osde.py` (basis functions, OSDE, leave-one-out, extended and spherical-harmonic bases), `summary.py`, `histo.py`, `style.py`.

`make_plots.py`, `plot_config.py` validation plots; `nu_pz_plot.py`, `lep_had_compare.py`, `hh_plots.py`, `acceptance_plots.py`.

`osde_fit.py` fits $f_{t\bar{t}}$ per level (`--basis paper|extended|sh`), applies it to $t\bar{t}$ and hh, projections and scores; `osde_simplify.py` (truncation, CV), `osde_order_scan.py`, `osde_truncation_check.py`, `osde_compare.py`, `basis_compare.py`, `paper_style_plots.py`.



N = 6 leading jets, no b-tag, nominal L1 inputs (pT 0.5 GeV, η 0.025, φ 2π/256, MET 1 GeV); (a) K = Kχ = 1024, (b) W = 18.
(a, b) ΔAUC(hh vs $t\bar{t}$) relative to the float L1 chain (AUC 0.583 full, 0.582 masses only; offline reference 0.602 / 0.600); error bars: paired bootstrap ±1σ. Shaded: recommended W = 16, K = 256.
(c, d) RMS of the event score vs the same-input float chain, events with the same χ^2 pick (full and masses-only coincide). Same χ^2 pick vs W = 10, 12, 14, 16, 18, 24: 32%, 70%, 89%, 96%, 97%, 97%.

Figure 38: AUC and per-event fidelity versus datapath width and ROM depth.

`ml_compare.py` the unsupervised and supervised machine-learning comparison.

`study_washout/` the mechanism and the step-by-step decomposition of the parton-to-gen loss.

`study_salvage/` the mass extension (`simplest.py`, `salvage.py`, `reco_extension.py`), the angular-term study (`angle_terms.py`), the truncation-noise test (`truncation_noise.py`) and the tagging study (`tagging.py`).

`study_bsm/`, `pileup/` the $H \to t\bar{t}$ benchmark and the pileup study.

`make_slides.py`, `note/make_note.py` the slides and this note; both read the same result files.

`tests/test_osde.py` 14 unit tests of the basis functions and the estimator (`pytest`).

`LOG.md` the running log of the study.

The slides are rebuilt with `python make_slides.py --sample samples/ttbar_bblvqq/run_seed12345 --hh samples/hh_bblvqq/run_seed2001 --osde osde --acceptance acceptance` and this note with `python note/make_note.py`.

Acknowledgement

The analysis code, the figures, the slides and this note were produced with Claude Code (Anthropic) under the direction of the author; every result was cross-checked as described in the text.